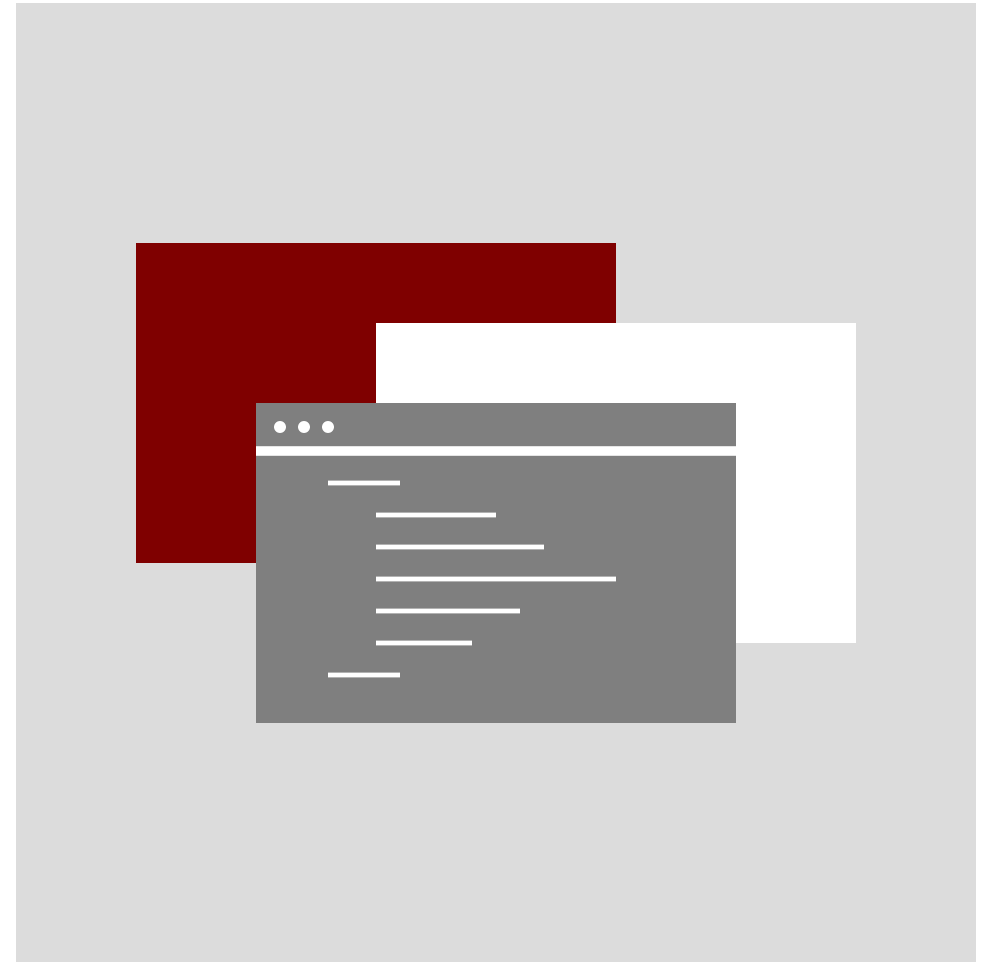


decksh

a little language for decks



Anthony Starks
@ajstarks

` A language is any mechanism to express intent,
and the input to many programs can be viewed
profitably as statements in a language. This column
is about those “little languages.”`

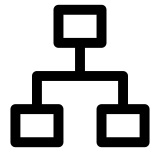
Jon Bentley, Little Languages, Communications of the ACM, August 1986

Deck



a Go package for presentations

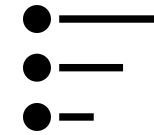
Elements



Structure



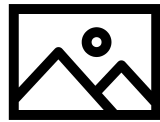
Text



Lists



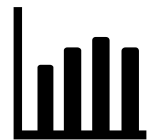
Arrows



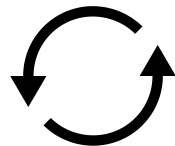
Images



Graphics



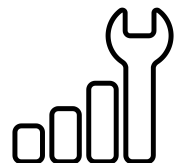
Charts



Loops



Data



Utility

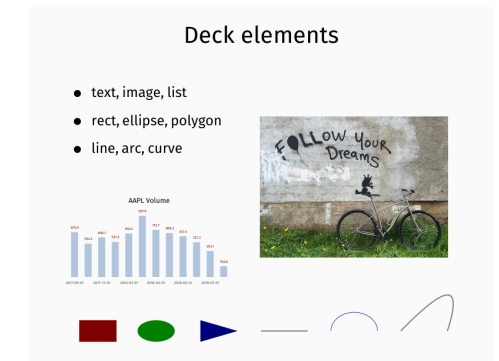
decksh → deck markup → SVG PDF PNG

```
deck
  slide "rgb(250,250,250)" "black"
    ctext "Deck elements" 50 90 5
    image "follow.jpg" 70 50 640 480 50
    blist 10 75 3
      li "text, image, list"
      li "rect, ellipse, polygon"
      li "line, arc, curve"
    elist

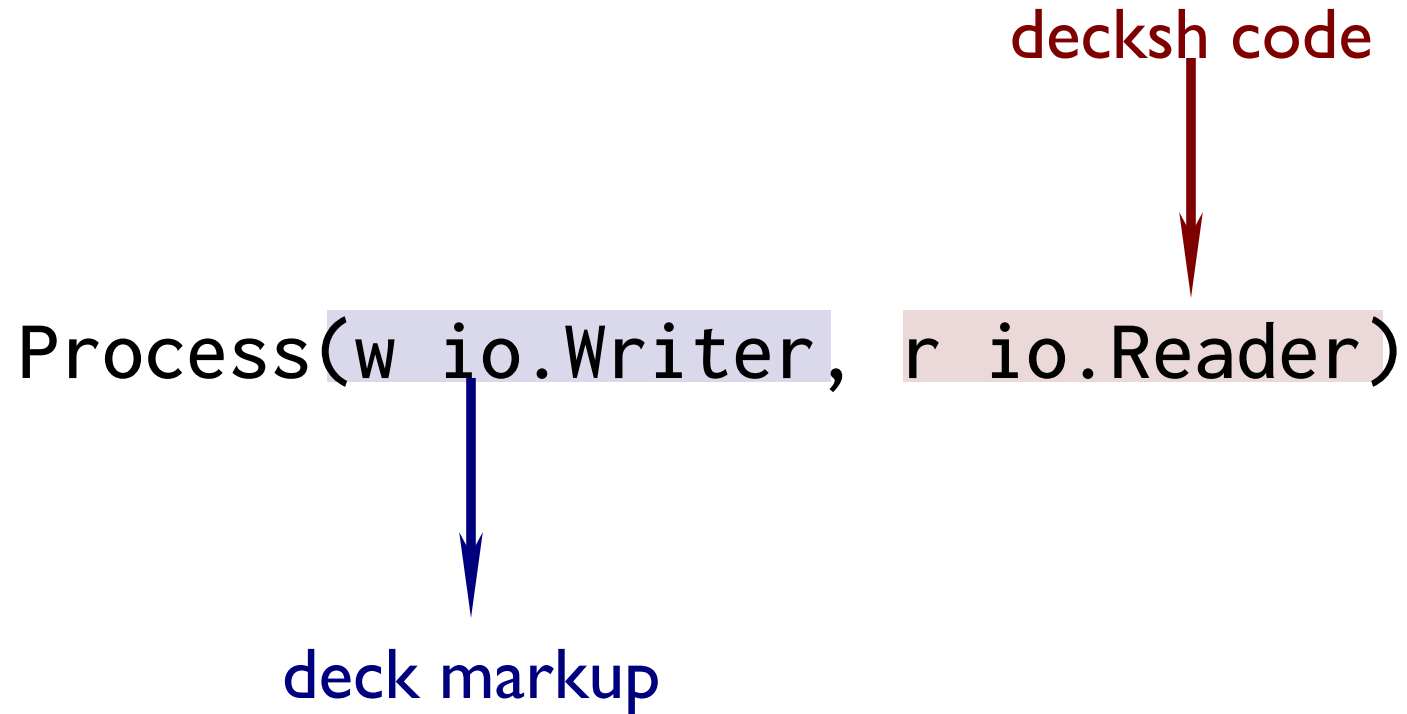
    gy=10
    rect 15 gy 8 6 "rgb(127,0,0)"
    ellipse 27.5 gy 8 6 "rgb(0,127,0)"
    line 50 gy 60 gy
    curve 80 gy 95 30 90 gy
    arc 70 gy 10 8 0 180 0.1 "rgb(0,0,127)"
    polygon "37 37 45" "13 7 10" "rgb(0,0,127)"

    opts="-fulldeck=f -textsize 1 -xlabel=2 -barwidth 1.5"
    dchart -left 10 -right 42 -top 42 -bottom 25 opts AAPL.d
  eslide
edeck
```

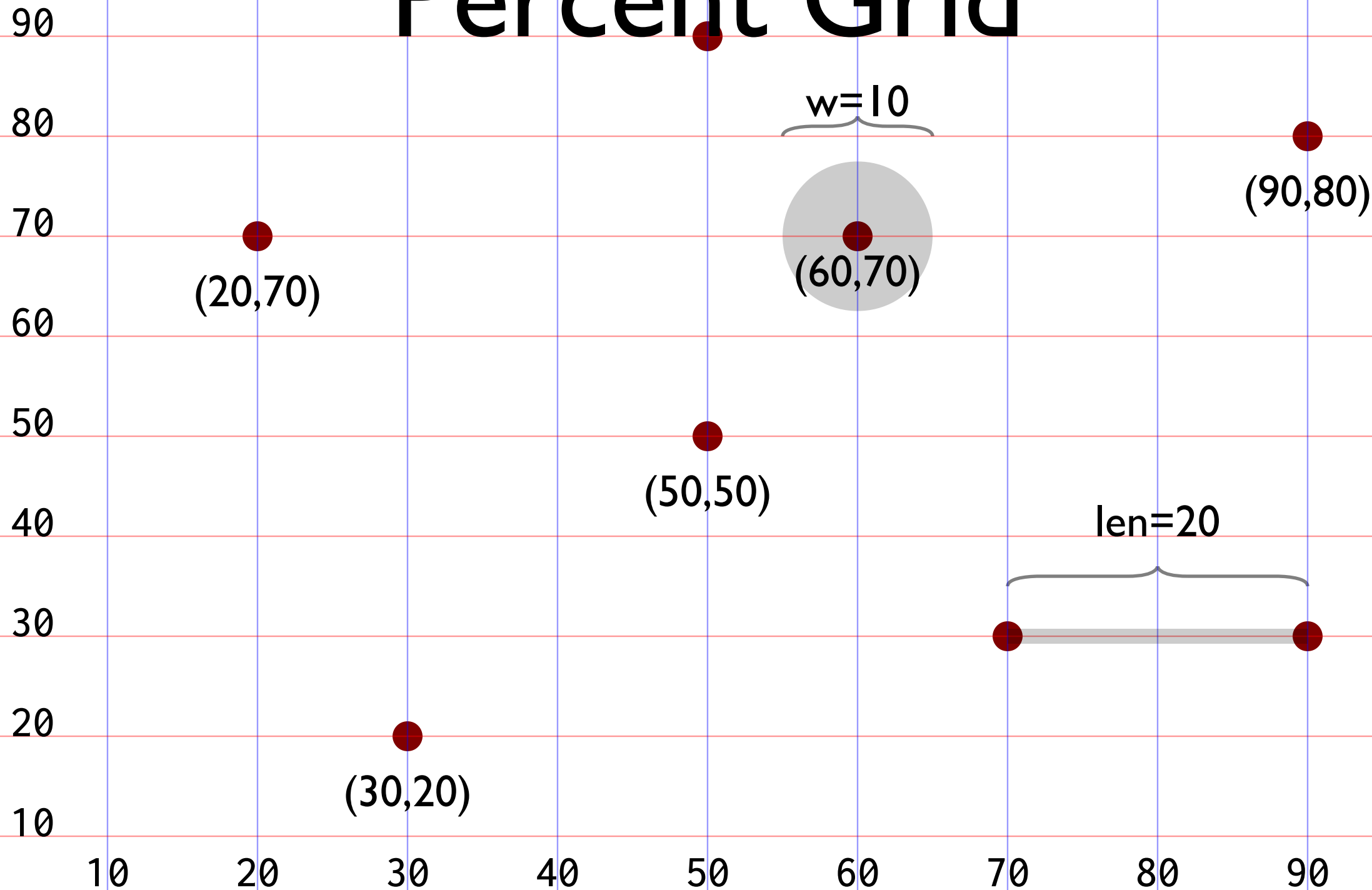
```
<deck>
<slide bg="rgb(250,250,250)" fg="black">
<text align="c" xp="50" yp="90" sp="5">Deck elements</text>
<image name="follow.jpg" xp="70" yp="50" width="640" height="480" scale="50" />
<list type="bullet" xp="10" yp="75" sp="3">
<li>text, image, list</li>
<li>rect, ellipse, polygon</li>
<li>line, arc, curve</li>
</list>
<rect xp="15" yp="10" wp="8" hp="6" color="rgb(127,0,0)" />
<ellipse xp="27.5" yp="10" wp="8" hp="6" color="rgb(0,127,0)" />
<line xp1="50" yp1="10" xp2="60" yp2="10" />
<curve xp1="80" yp1="10" xp2="95" yp2="30" xp3="90" yp3="10" />
<arc xp="70" yp="10" wp="10" hp="8" a1="0" a2="180" sp="0.1" color="rgb(0,0,127)" />
<polygon xc="37 37 45" yc="13 7 10" color="rgb(0,0,127)" />
<text xp="26.00" yp="45.60" sp="1.50" align="center" wp="0.00" font="sans" opacity="100.00"
color="black" type="">AAPL Volume</text>
<line xp1="10.00" yp1="25.00" xp2="10.00" yp2="37.46" sp="1.50" opacity="100.00"
color="lightsteelblue" />
<text xp="10.00" yp="38.46" sp="0.75" align="center" wp="0.00" font="sans" opacity="100.00"
color="rgb(127,0,0)" type="">679.9</text>
<text xp="10.00" yp="23.00" sp="0.80" align="center" wp="0.00" font="sans" opacity="100.00"
color="rgb(75,75,75)" type="">2017-09-01</text>
<line xp1="12.91" yp1="25.00" xp2="12.91" yp2="34.24" sp="1.50" opacity="100.00"
color="lightsteelblue" />
<text xp="12.91" yp="35.24" sp="0.75" align="center" wp="0.00" font="sans" opacity="100.00"
color="rgb(127,0,0)" type="">504.3</text>
...
</slide>
</deck>
```



decksh API



Percent Grid



```
// hello world
deck
    slide "black" "white"
        ctext "hello, world" 50 25 10
        circle 50 0 100 "blue"
    eslide
edeck
```

hello, world

Running decksh

`decksh`

read from stdin, write to stdout

`decksh in.dsh`

read from file, write to stdout

`decksh -o out.xml`

read from stdin, write to file

`decksh -o out.xml in.dsh`

read from file, write to file

`chmod +x in.dsh; ./in.dsh`

executable deck with `#!/path/to/decksh`

`decksh example.dsh | pdfdeck ...`

hw.dsh - Visual Studio Code

File Edit Selection View Go Debug Terminal Help

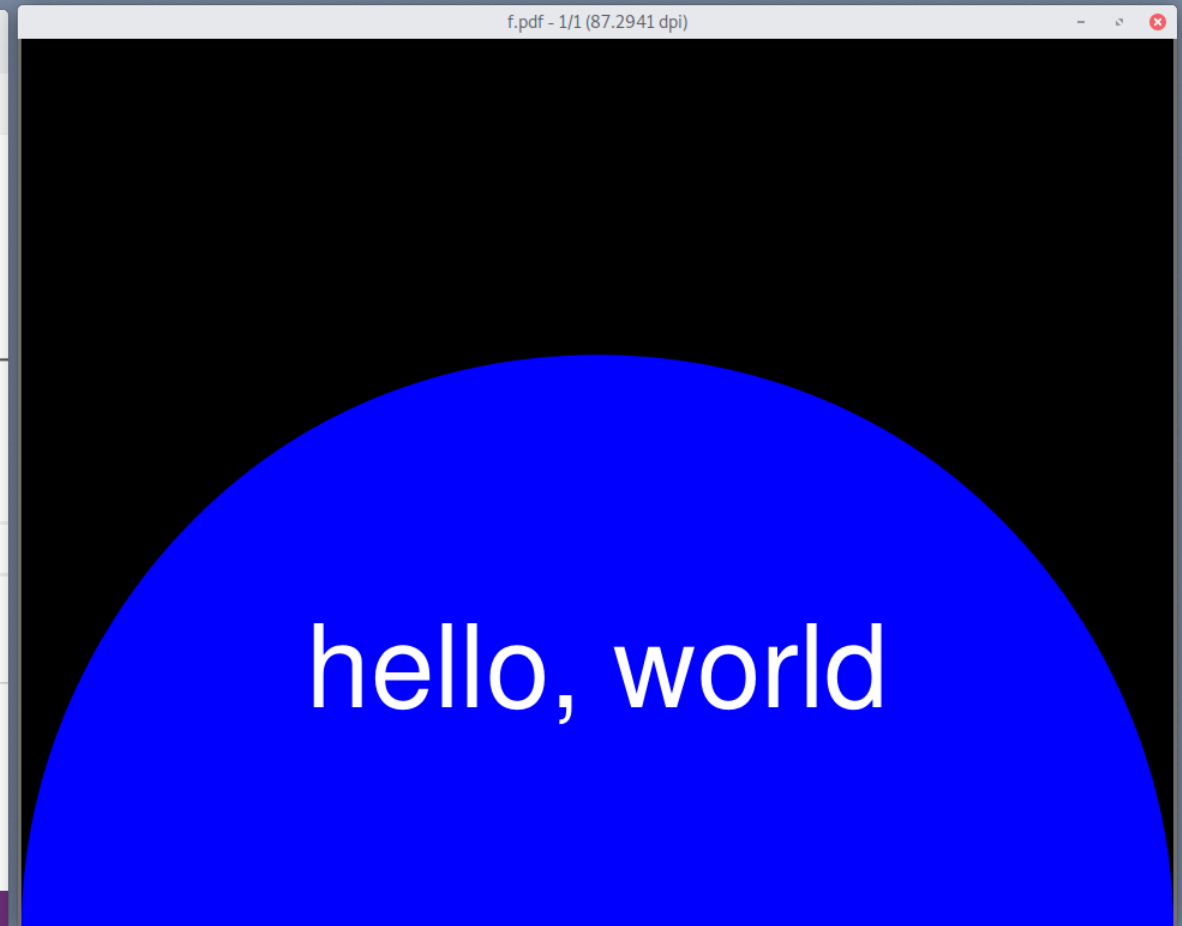
hw.dsh x

// hello world
deck
 slide "black" "white"
 ctext "hello, world" 50 25 10
 circle 50 0 100 "blue"
 eslide
edeck

PROBLEMS TERMINAL ... 1: bash

\$ decksh hw.dsh | pdf
\$ open f.pdf
\$

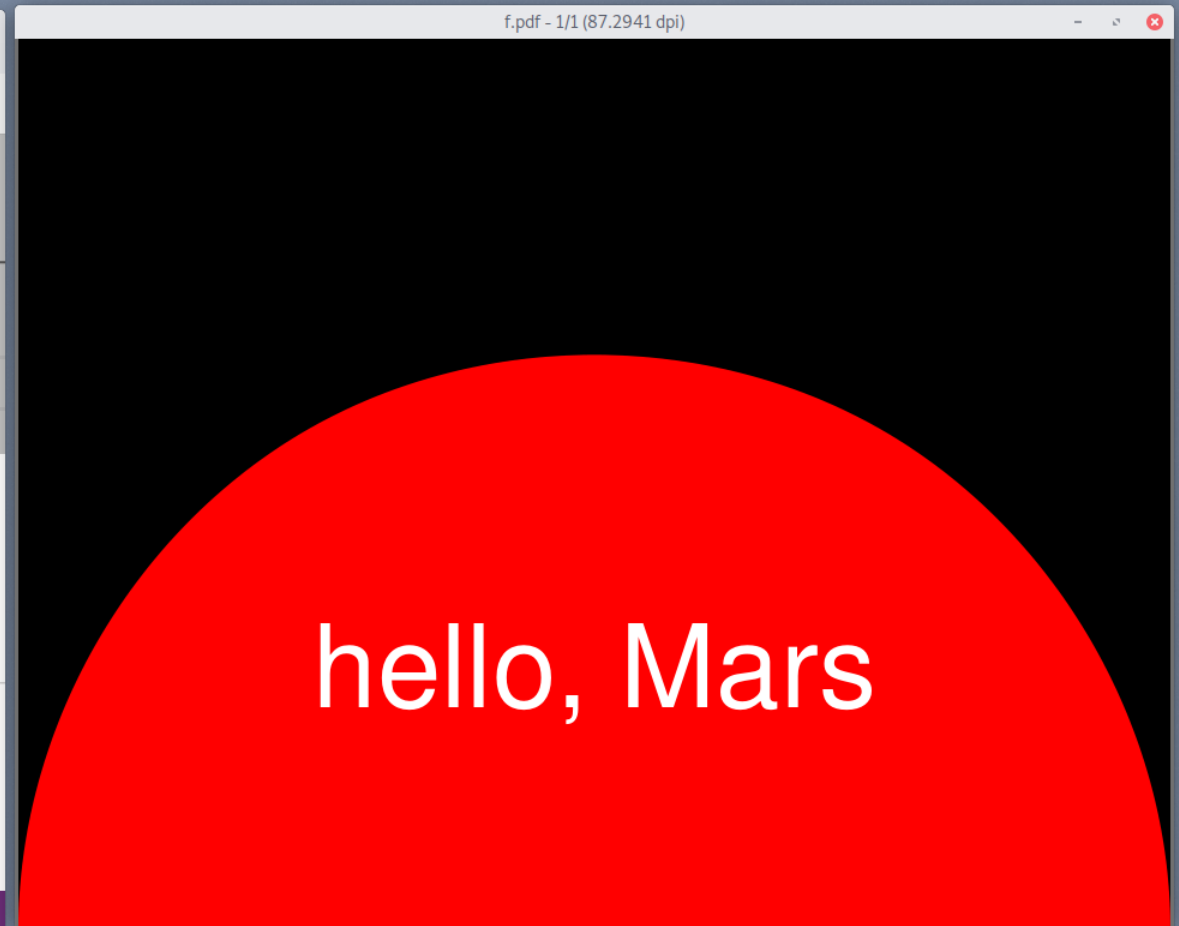
master* 0 0 Ln 8, Col 1 Tab Size: 4 UTF-8 LF Plain Text



```
hw.dsh - Visual Studio Code
File Edit Selection View Go Debug Terminal Help

hw.dsh x
// hello world
deck
  slide "black" "white"
    ctext "hello, Mars" 50 25 10
    circle 50 0 100 "red"
  eslide
edeck

PROBLEMS TERMINAL ...
1: bash
$ decksh hw.dsh | pdf
$ open f.pdf
$ decksh hw.dsh | pdf
$
```



Keywords and arguments

keyword

arguments

mandatory

optional

text `"..string...." x y n`

`"font" "color" op`

text "hello, world" 80 50 2

hello, world

text "hello, world" 80 40 2

"serif"

hello, world

text "hello, world" 80 30 2

"serif" "red"

hello, world

text "hello, world" 80 20 2

"serif" "red" 50

hello, world

Variables and Assignments

```
x=10 // number assignment
y=20
factor=2
what="hello world" // string assignment

size=x/factor // assignment with binop
text what x y size // text "hello world" 10 20 5

y-=10 // assignment operation
size+=factor // assignment op, substitute
text what x y size // text "hello world" 10 10 7

for v=0 100 5 // loop from 0 to 100 by 5
    line 100 v 0 v 0.1 "blue" // blue horizontal lines
    line v 100 v 0 0.1 "red" // red vertical lines
efor
```

Keywords

Structure Text

deck/edeck
slide/eslide
canvas
def/edef
for/efor
func
grid
import
include
if/else/eif
text
btext
ctext
etext
para/epara
rtext
arctext
textblock
textblockfile
textfile
textcode

Graphics

acircle
arc
circle
curve
ellipse
dline
hline
line
pill
polygon
polyline
rect
rrect
ruler
square
star
vline

Lists

list
blist
nlist
clist
li
elist

Braces

lbrace
rbrace
ubrace
dbrace
lbracket
rbracket
dbracket
ubracket

Arrows

arrow
rcarrow
lcarrow
ucarrow
dcarrow

Maps

geoarc
geoborder
geobound
geocanvas
geoimage
geolabel
geoloc
geomark
geopath
geopathfile
georegion
georuler

Images

image
cimage

Charts

chartbbox
dchart
legend
area
barchart
hbar
wbar
linechart
scatter
slopechart
donut
pie
waffle
pgrid
pmap
bowtie
fanchart

Assign

polar
polarx
polary
random
area
format
substr
vmap
dump

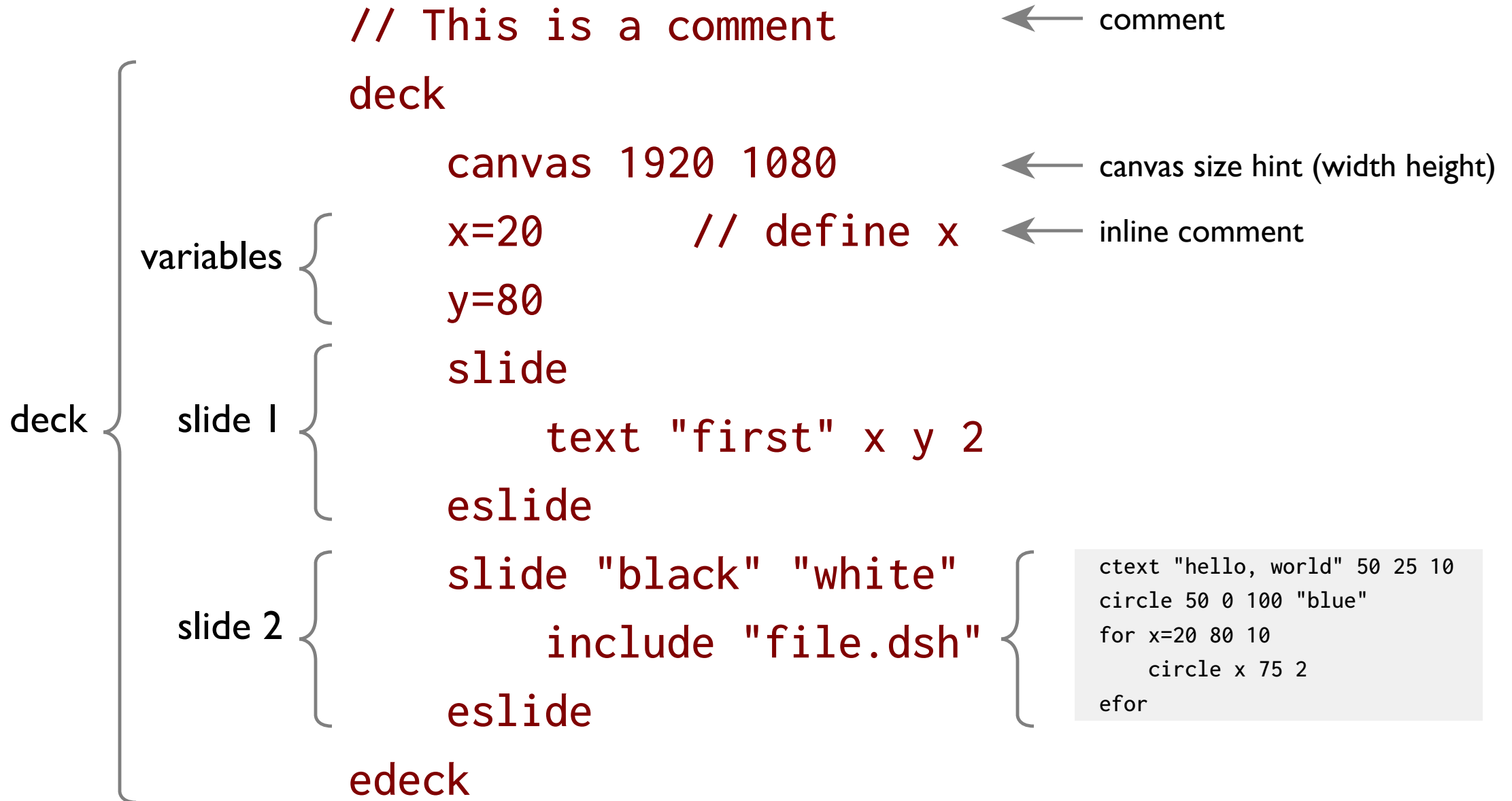
Math

geocoord
cosine
sine
sqrt
tangent

Data

data
edata
content

Structure



Text

hello world

text

x y size [font] [color] [op] [link]

hello world

ctext

x y size [font] [color] [op] [link]

hello world.

etext

x y size [font] [color] [op] [link]

hello world
there world

rtext

x y angle size [font] [color] [op] [link]

hello there
world

arctext

cx cy radius beg-angle end-angle size [font] [color] [op] [link]

Text

textblock

The quick brown fox
jump over the lazy
dog

"text" x y width size [font] [color] [op] [link]

textfile

This is the contents
of a file. it has lines of text.
Reading is fundamental.

"filename" x y size [font] [color] [op] [sp]

textcode

```
import "fmt"
func main() {
    fmt.Println("Go")
}
```

"filename" x y width size [color]

Lists

First thing

Second thing

Third thing

Fourth

- First thing

- Second thing

- Third thing

- Fourth

1. First thing

2. Second thing

3. Third thing

4. Fourth

First thing

Second thing

Third thing

Fourth

`list`

`li "..."`

`elist`

`blist`

`li "..."`

`elist`

`nlist`

`li "..."`

`elist`

`clist`

`li "..."`

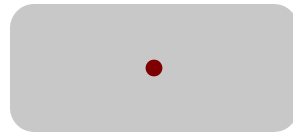
`elist`

Graphics (shapes)



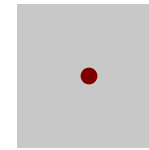
rect

x y w h [color] [op]



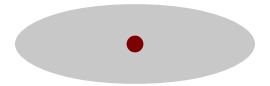
rrect

x y w h r [color]



square

x y w [color] [op]



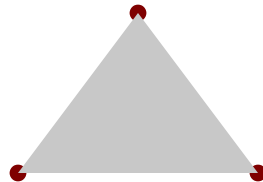
ellipse

x y w h [color] [op]



circle

x y w [color] [op]



polygon

"xc" "yc" [color] [op]



pill

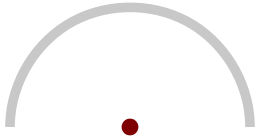
x y w h [color]



star

x y nsides in out [color] [op]

Graphics (lines)



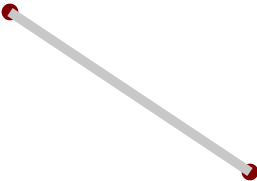
arc

x y w h a1 a2 [lw] [color] [op]



curve

bx by cx cy ex ey [lw] [color] [op]



line

x1 y2 x2 y2 [lw] [color] [op]



hline

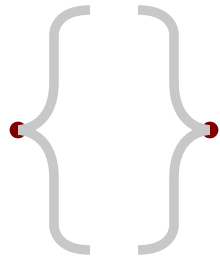
x y len [lw] [color] [op]



vline

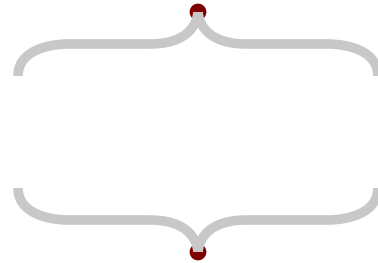
x y len [lw] [color] [op]

Braces and Brackets



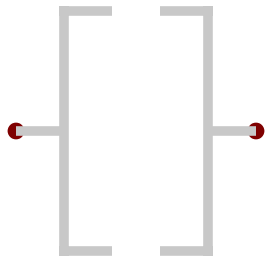
[l-r]brace

x y size aw ah [lw] [color] [op]



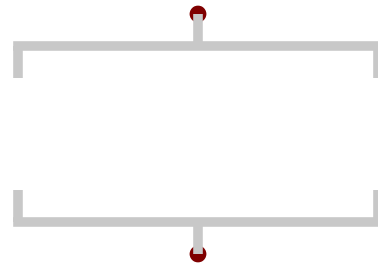
[u-d]brace

x y size aw ah [lw] [color] [op]



[l-r]bracket

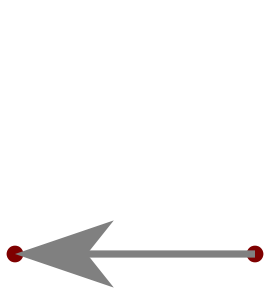
x y w h [lw] [color] [op]



[u-d]bracket

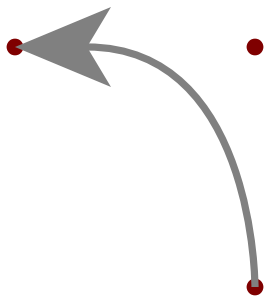
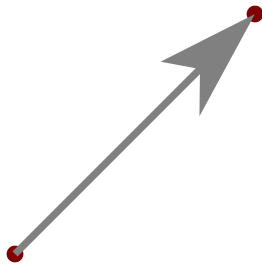
x y w h [lw] [color] [op]

Arrows

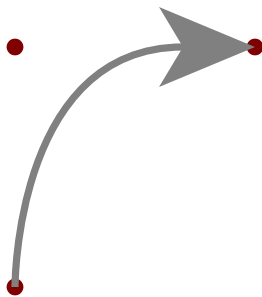


arrow

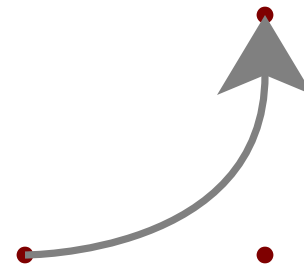
x1 y1 x2 y2 [linewidth] [aw] [ah] [color] [op]



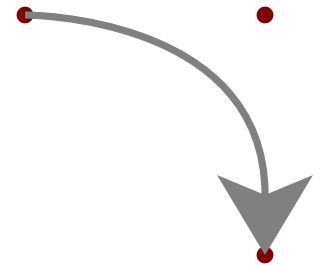
lcarrow



rcarrow



ucarrow



dcarrow

x1 y1 x2 y2 x3 y3 [lw] [aw] [ah] [color] [op]

...

...

...

Images



image

"filename" x y w h [scale] [link]



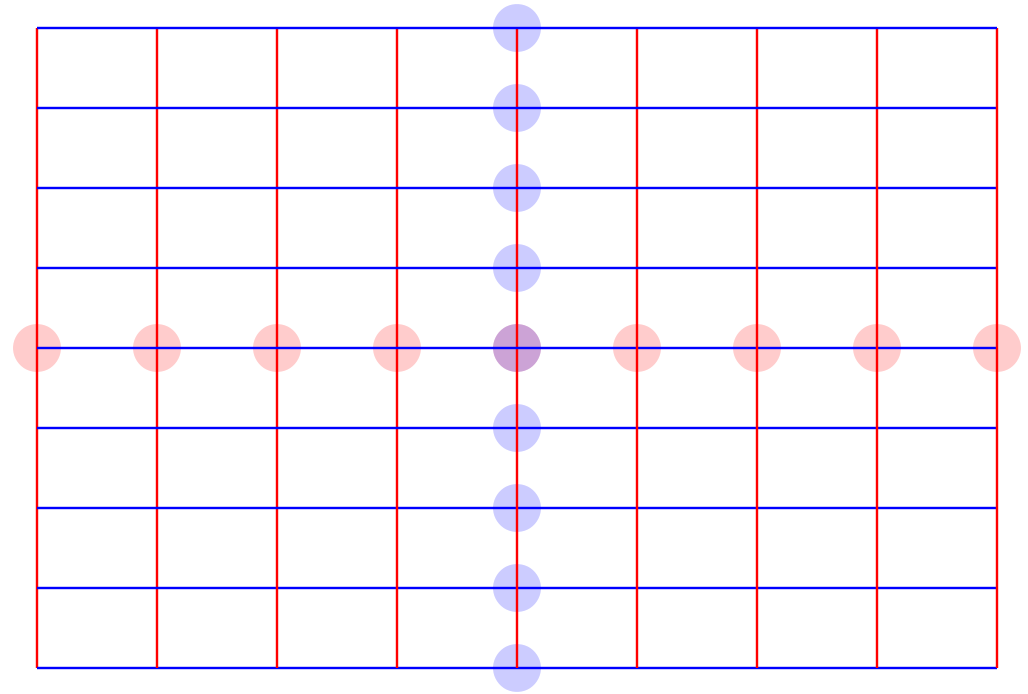
Up in the clouds

cimage

"filename" "caption" x y w h [scale] [link] [caption-size]

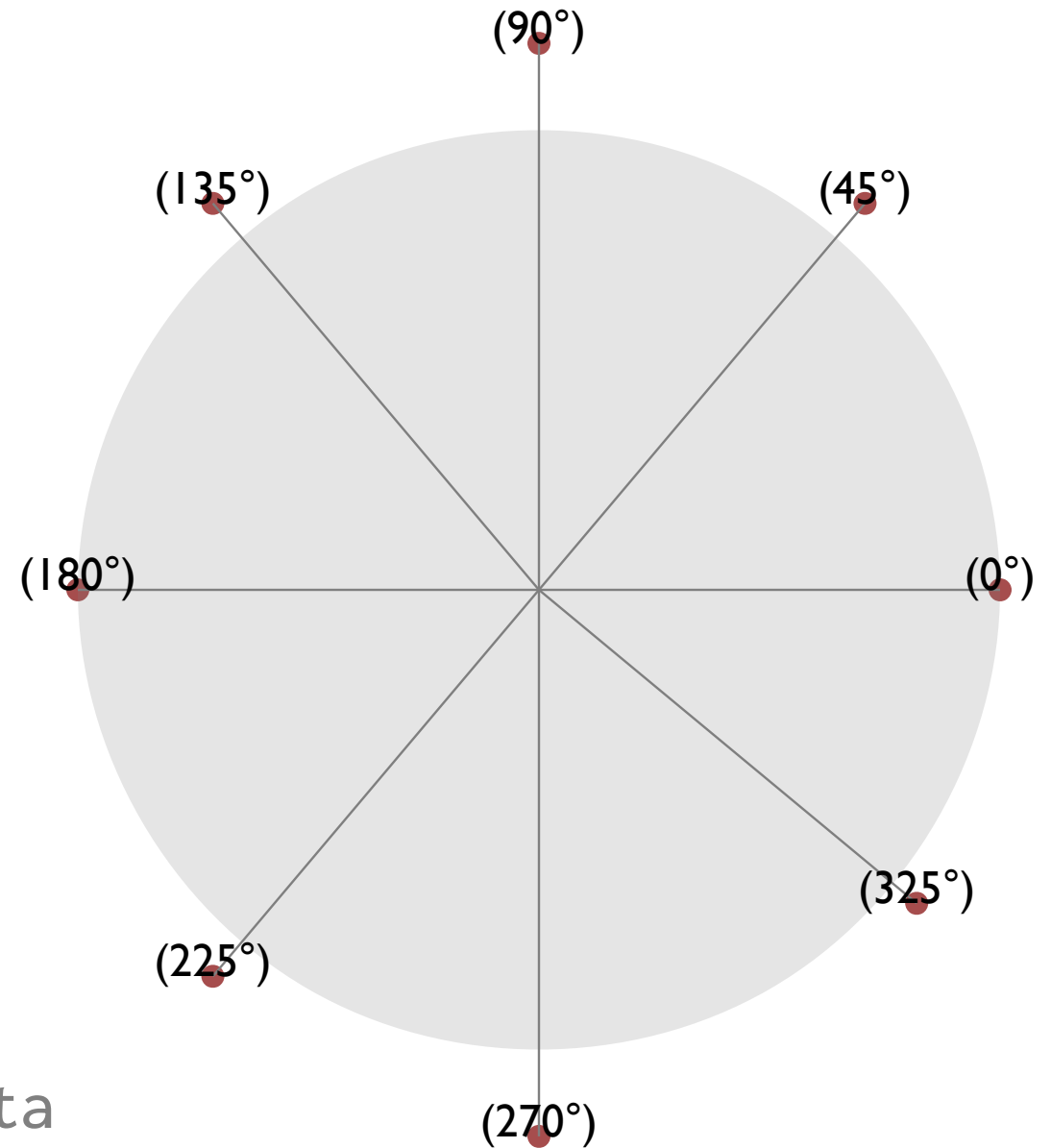
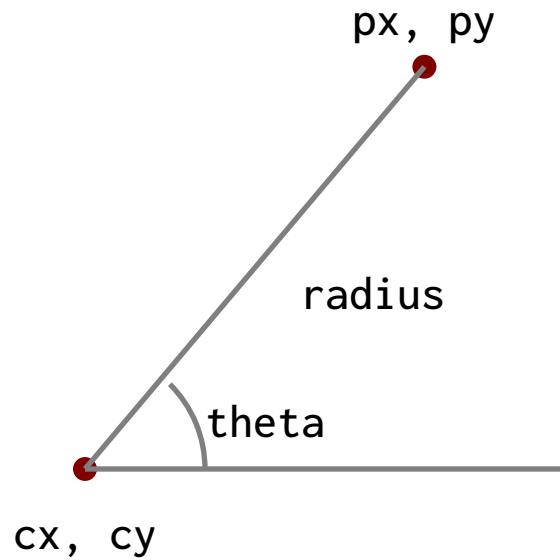
Loops

```
for v=50 90 5
    vline v 50 40 0.1 "red"
    hline 50 v 40 0.1 "blue"
    circle v 70 2 "red" 20
    circle 70 v 2 "blue" 20
efor
```



```
for v=begin end [increment]
    ...v...
efor
```

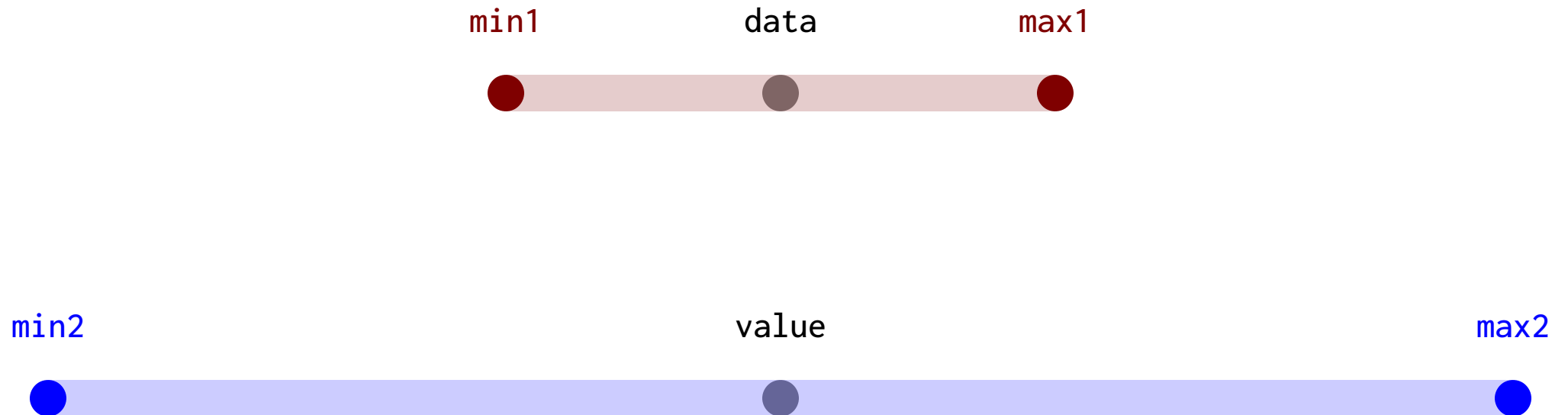
Polar Coordinates



$px = \text{polarx}$ cx cy $radius$ $theta$

$py = \text{polary}$ cx cy $radius$ $theta$

Mapping Ranges



```
value= vmap data min1 max1 min2 max2
```

Formatted Strings

```
v1=100.3  
v2=200.234  
title=format "%.2f Million (USD)" v1  
subtitle=format "Total value: %.2f" v1+v2
```

```
ctext title      80 70 4 "sans" "maroon"  
ctext subtitle  80 60 3 "sans" "gray"
```

100.30 Million (USD)

Total value: 300.53

```
value= format  fmt expression
```

Random Numbers

```
x1=random 40 70
```

```
y1=random 60 70
```

x1,y1



```
x2=random 40 50
```

```
y2=random 50 60
```

x2,y2



```
x3=random 60 70
```

```
y3=random 35 45
```

x3,y3



```
value= random min max
```

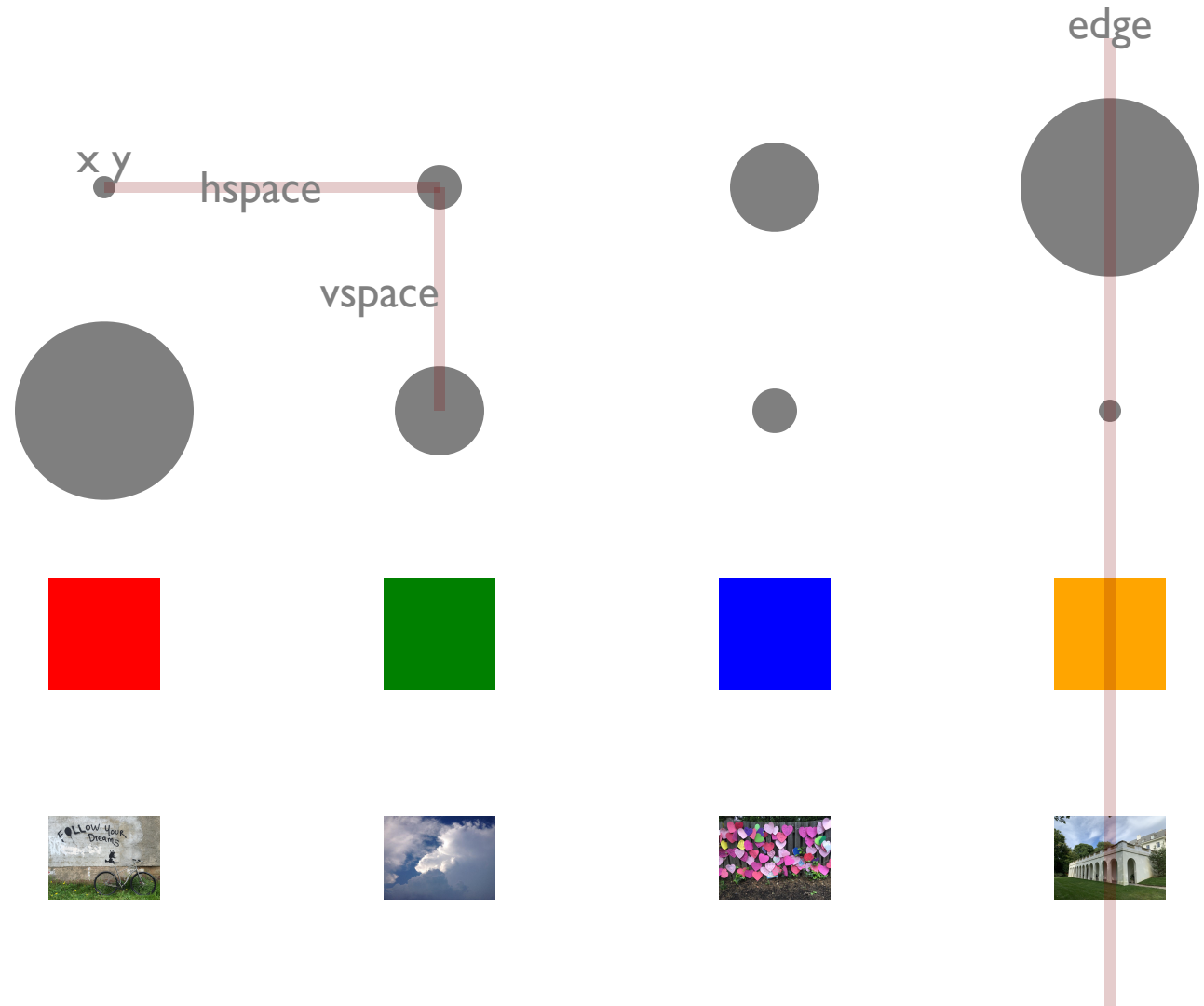
Flexible Grid

```
circle x y 1  
circle x y 2  
circle x y 4  
circle x y 8
```

```
circle x y 8  
circle x y 4  
circle x y 2  
circle x y 1
```

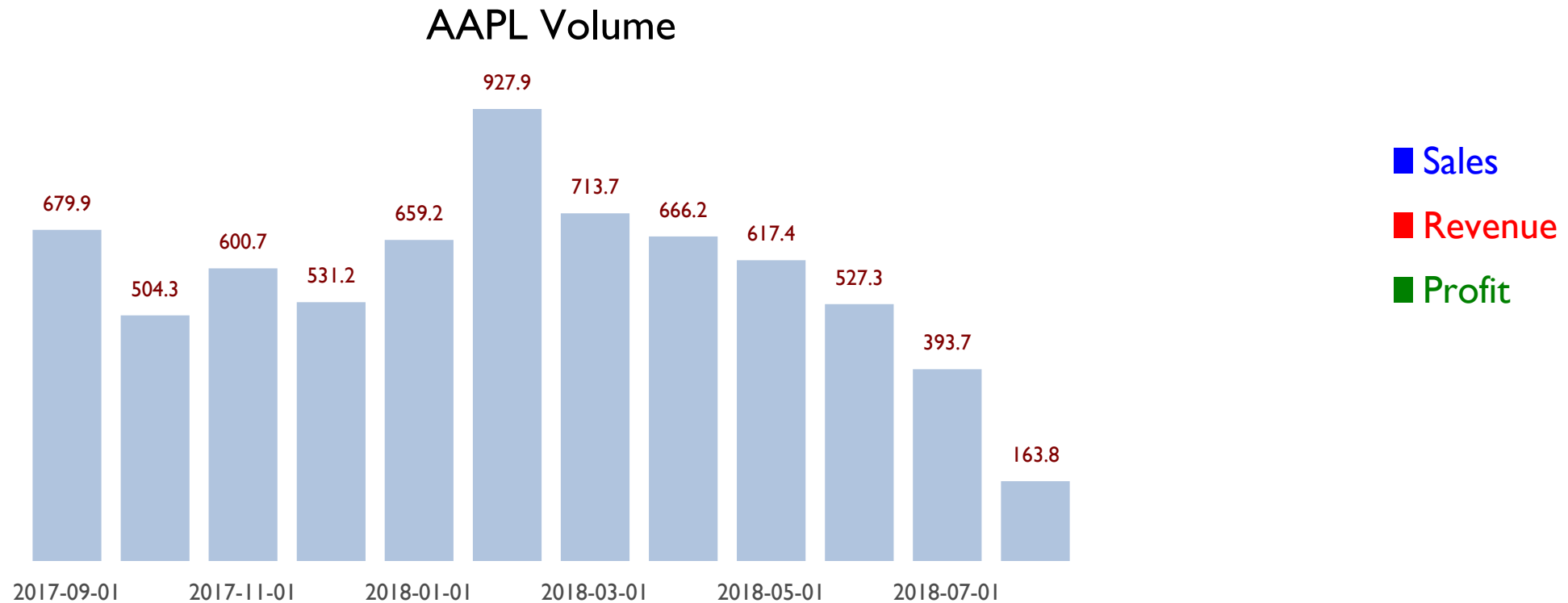
```
square x y 5 "red"  
square x y 5 "green"  
square x y 5 "blue"  
square x y 5 "orange"
```

```
image "images/follow.jpg" x y 640 480 7  
image "images/cloudy.jpg" x y 640 480 7  
image "images/hearts.jpg" x y 640 480 7  
image "images/oldfields.jpg" x y 640 480 7
```



grid "foo.dsh" x y hspace vspace edge

Charts



dchart

[args]

legend

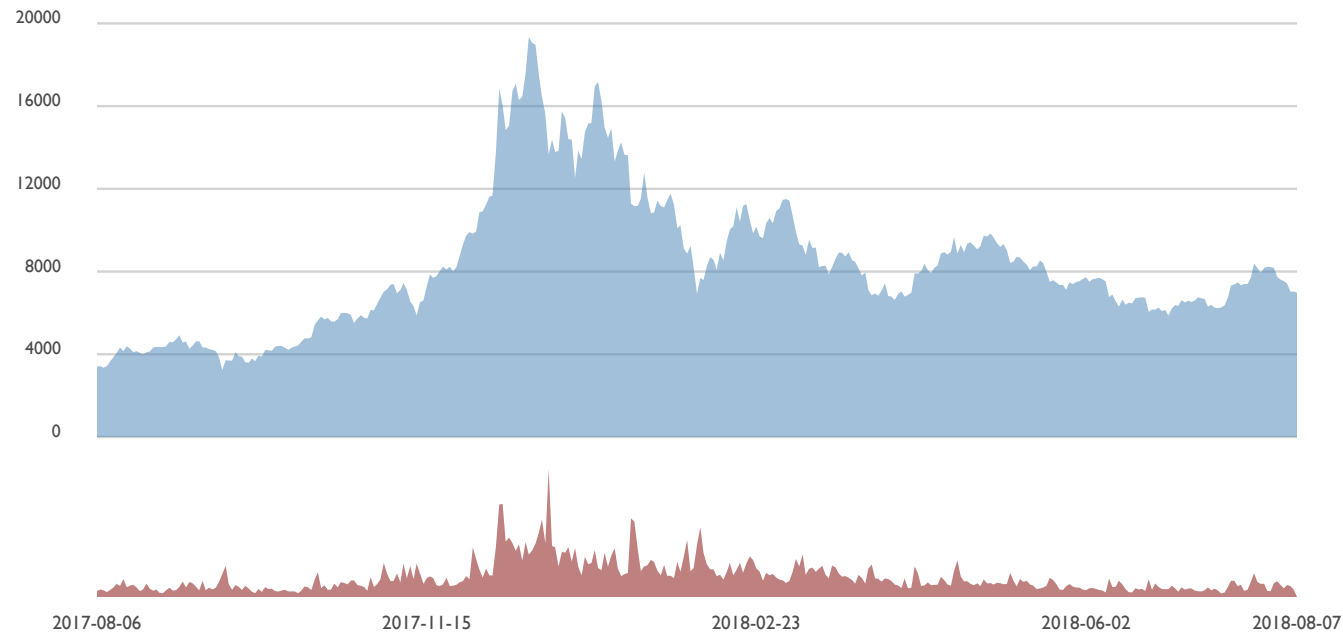
x y size [font] [color]

dchart options: chart types

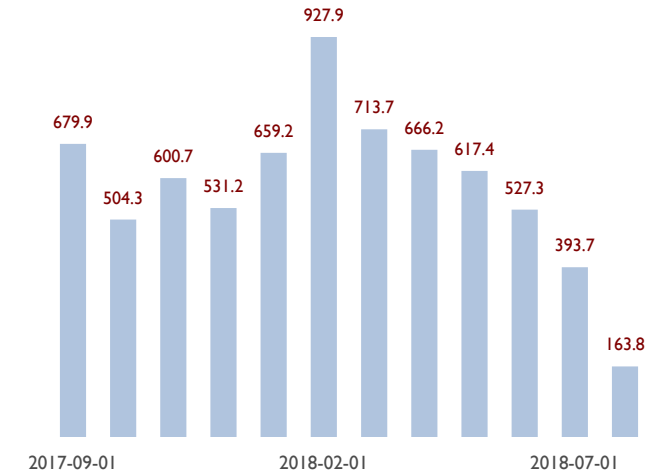
option	default	description
-bar	true	bar chart
-wbar	false	word bar chart
-hbar	false	horizontal bar chart
-donut	false	donut chart
-dot	false	dot chart
-lego	false	lego chart
-line	false	line chart
-pgrid	false	proportional grid
-pmap	false	proportional map
-bowtie	false	bowtie chart
-fan	false	fan chart
-radial	false	radial chart
-scatter	false	scatter chart
-slope	false	slope chart
-vol	false	volume (area) chart

dchart: charts for deck

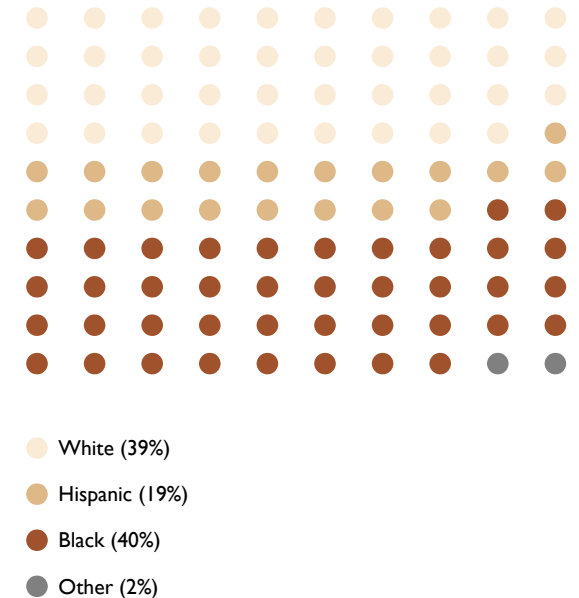
Bitcoin to USD



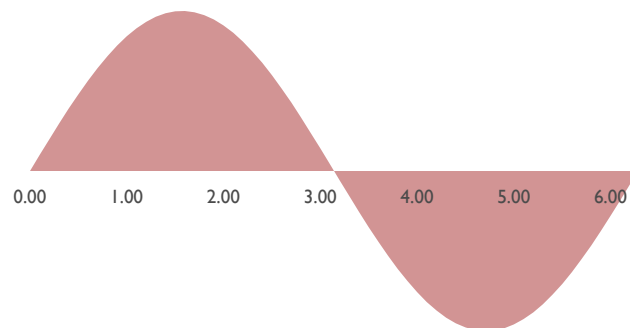
AAPL Volume



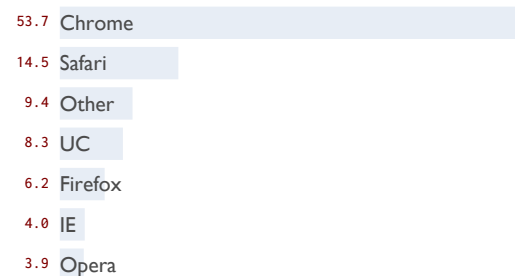
US Incarceration Rate



$y=\sin(x)$



Browser Market Share Dec 2016-Dec 2017



deck

```
slide "rgb(250,250,250)" "black"
```

```
  ctext  "Deck elements" 50 90 5
```

```
  image  "follow.jpg"    70 50 640 480 50
```

```
  blist  10 75 3
```

```
    li "text, image, list"
```

```
    li "rect, ellipse, polygon"
```

```
    li "line, arc, curve"
```

```
  elist
```

```
  gy=10
```

```
  rect    15 gy 8 6          "rgb(127,0,0)"
```

```
  ellipse 27.5 gy 8 6        "rgb(0,127,0)"
```

```
  line    50 gy 60 gy
```

```
  curve   80 gy 95 30 90 gy
```

```
  arc     70 gy 10 8 0 180 0.1 "rgb(0,0,127)"
```

```
  polygon "37 37 45" "13 7 10" "rgb(0,0,127)"
```

```
  opts="-fulldeck=f -textsize 1 -xlabel=2 -barwidth 1.5"
```

```
  dchart -left 10 -right 42 -top 42 -bottom 25 opts AAPL.d
```

```
eslide
```

edek

Deck elements

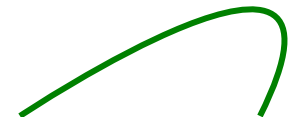
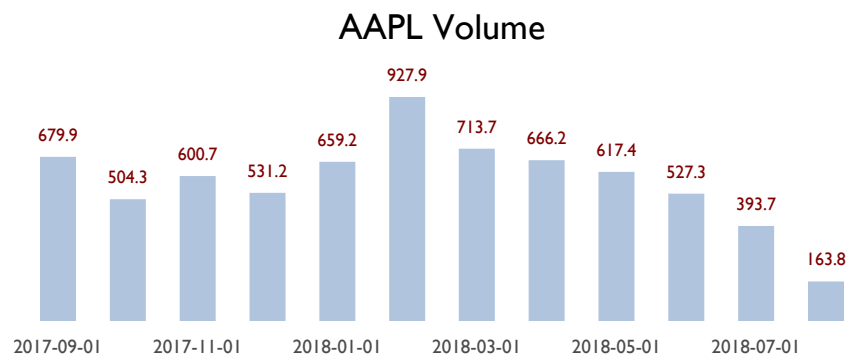
- text, image, list
- rect, ellipse, polygon
- line, arc, curve



decksh example.dsh | pdf

Deck elements

- text, image, list
- rect, ellipse, polygon
- line, arc, curve



text

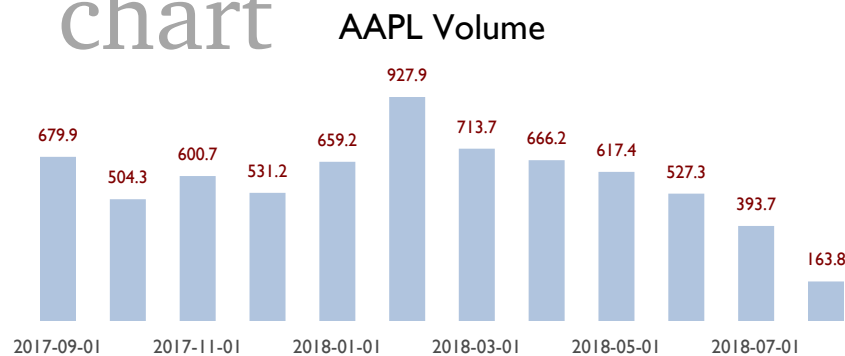
Deck elements

list

image

- text, image, list
- rect, ellipse, polygon
- line, arc, curve

chart



rect



ellipse



polygon



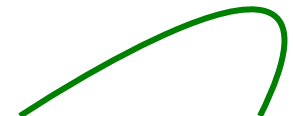
line



arc



curve



Examples



Anthony J. Starks

Art + Code



+1 908.548.3403



ajstarks@gmail.com



@ajstarks



github.com/ajstarks



speakerdeck.com/ajstarks

```

deck
  mx=50          // midpoint
  tx=30          // text left
  ix=20          // image left
  ts=10          // base text size
  ss=ts*0.85     // sub-head text size
  cs=ts*0.55     // contact info text size
  ly=58          // line y

slide "white" "rgb(100,100,100)"
  image "starx.png"          mx 87 512 512 7.5
  ctext "Anthony J. Starks"  mx 70 ts "sans" "black"
  ctext "Art + Code"         mx 62 ss "sans" "maroon"
  line  ix ly 80 ly 0.3 "maroon"

  image "phone.png"          ix 50 1200 1200 1.2
  image "email.png"          ix 40 1200 1200 1.2
  image "twitter.png"        ix 30 1200 1200 1.2
  image "github.png"         ix 20 120 120 10
  image "sd.png"             ix 10 512 512 2.5

  text "+1 908.548.3403"      tx 49 cs
  text "ajstarks@gmail.com"  tx 39 cs
  text "@ajstarks"           tx 29 cs
  text "github.com/ajstarks" tx 19 cs
  text "speakerdeck.com/ajstarks" tx 9 cs
eslide
edeck

```



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```

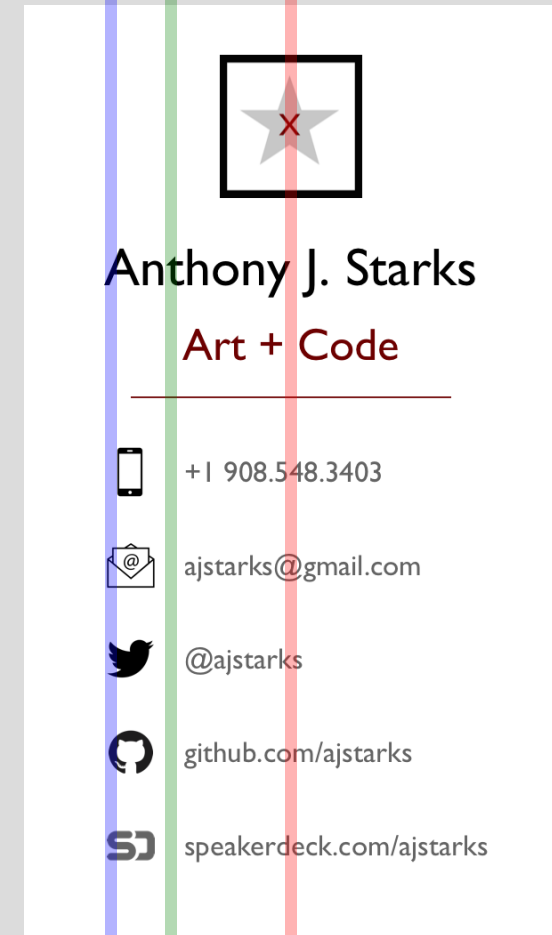
deck
  mx=50      // midpoint
  tx=30      // text left
  ix=20      // image left
  ts=10      // base text size
  ss=ts*0.85 // sub-head text size
  cs=ts*0.55 // contact info text size
  ly=58      // line y

  slide "white" "rgb(100,100,100)"
    image "starx.png"      mx 87 512 512 7.5
    ctext "Anthony J. Starks" mx 70 ts "sans" "black"
    ctext "Art + Code"      mx 62 ss "sans" "maroon"
    line  ix ly 80 ly 0.3 "maroon"

    image "phone.png"      ix 50 1200 1200 1.2
    image "email.png"      ix 40 1200 1200 1.2
    image "twitter.png"    ix 30 1200 1200 1.2
    image "github.png"     ix 20 120 120 10
    image "sd.png"         ix 10 512 512 2.5

    text "+1 908.548.3403"  tx 49 cs
    text "ajstarks@gmail.com" tx 39 cs
    text "@ajstarks"       tx 29 cs
    text "github.com/ajstarks" tx 19 cs
    text "speakerdeck.com/ajstarks" tx 9 cs
  eslide
edeck

```



```

deck
  mx=50          // midpoint
  tx=30          // text left
  ix=20          // image left
  ts=10          // base text size
  ss=ts*0.85     // sub-head text size
  cs=ts*0.55     // contact info text size
  ly=58          // line y

slide "white" "rgb(100,100,100)"
  image "starx.png"          mx 87 512 512 7.5
  ctext "Anthony J. Starks"  mx 70 ts "sans" "black"
  ctext "Art + Code"         mx 62 ss "sans" "maroon"
  line  ix ly 80 ly 0.3 "maroon"

  image "phone.png"          ix 50 1200 1200 1.2
  image "email.png"          ix 40 1200 1200 1.2
  image "twitter.png"        ix 30 1200 1200 1.2
  image "github.png"         ix 20 120 120 10
  image "sd.png"             ix 10 512 512 2.5

  text "+1 908.548.3403"      tx 49 cs
  text "ajstarks@gmail.com"  tx 39 cs
  text "@ajstarks"           tx 29 cs
  text "github.com/ajstarks" tx 19 cs
  text "speakerdeck.com/ajstarks" tx 9  cs
eslide
edeck

```



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speakerdeck.com/ajstarks

```

deck
  mx=25      // midpoint
  tx=62      // text left
  ix=57      // image left
  ts=6       // base text size
  ss=ts*0.85 // sub-head text size
  cs=ts*0.50 // contact info text size
  lx=50      // line x

  slide "white" "rgb(100,100,100)"
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    ctext "Anthony J. Starks" mx 35 ts "sans" "black"
    ctext "Art + Code"      mx 22 ss "sans" "maroon"
    line  lx 90 lx 10 0.3 "maroon"

    image "phone.png"      ix 80 1200 1200 1.2
    image "email.png"      ix 65 1200 1200 1.2
    image "twitter.png"    ix 50 1200 1200 1.2
    image "github.png"     ix 35 120 120 10
    image "sd.png"         ix 20 512 512 2.5

    text "+1 908.548.3403"  tx 79 cs
    text "ajstarks@gmail.com" tx 64 cs
    text "@ajstarks"       tx 49 cs
    text "github.com/ajstarks" tx 34 cs
    text "speakerdeck.com/ajstarks" tx 19 cs
  eslide
edeck

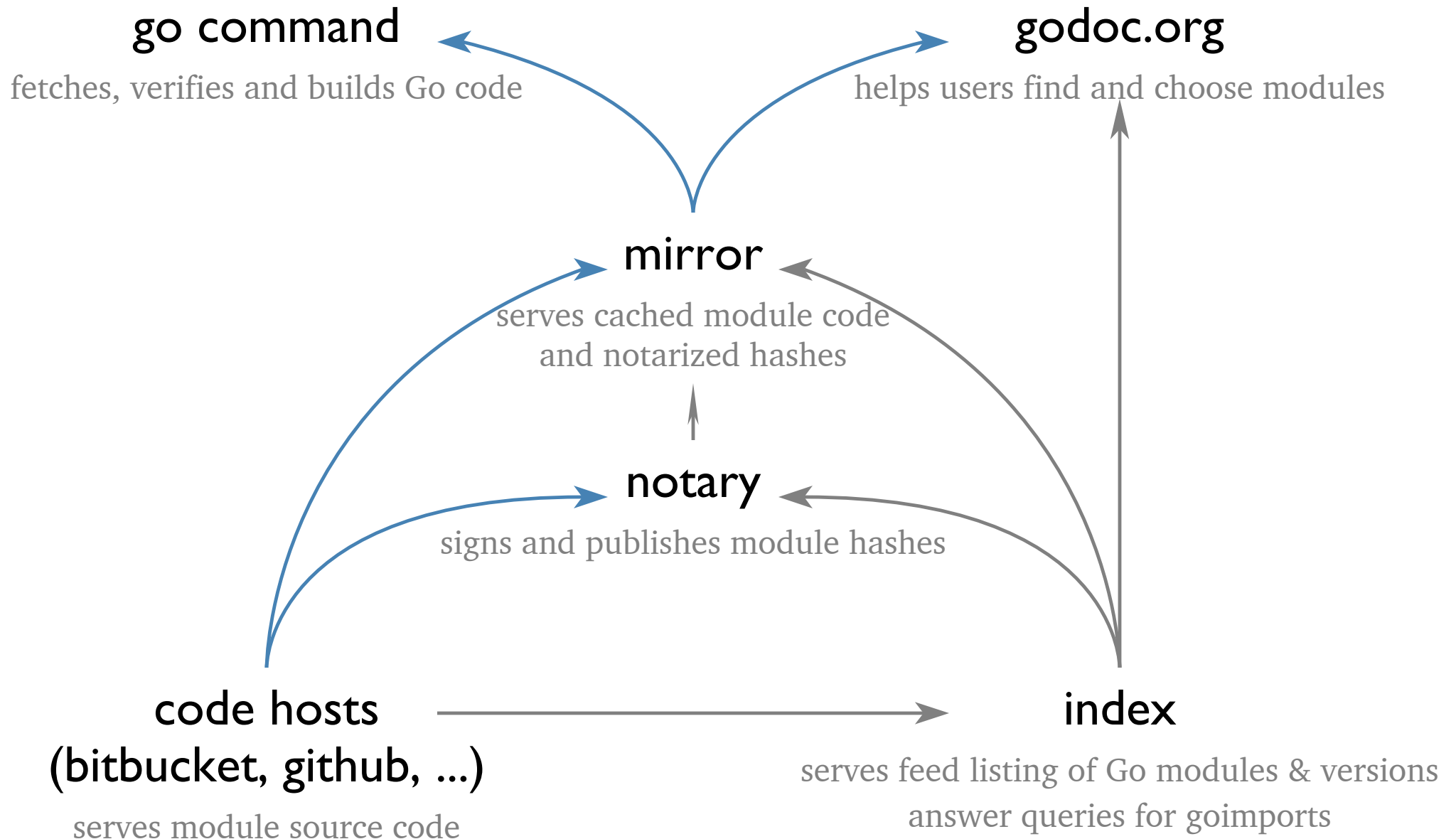
```



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 @ajstarks
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 speakerdeck.com/ajstarks

Go Module Information Flows



BOS



SFO

Virgin America 351

Gate B38

8:35am

On Time

JFK



IND

US Airways 1207

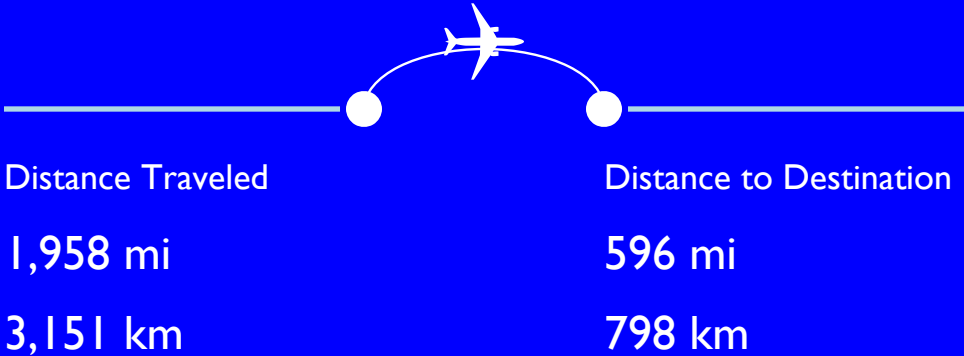
Gate C31C

5:35pm

Delayed

Flight Information

Los Angeles (LAX)  New York/Newark (EWR)



Ground speed



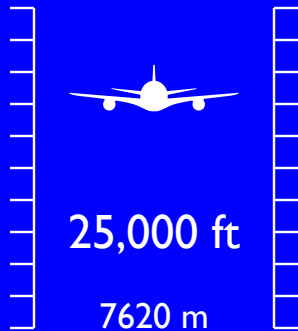
Headwind



Outside Temperature



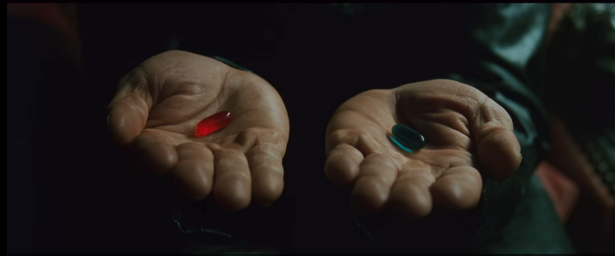
Current Altitude





Pulp Fiction (1993)

1



The Matrix (1999)

2



Roma (2018)

4

pulp04.png,1920,1080,Pulp Fiction (1993)

matrix12.png,1920,1080,The Matrix (1999)

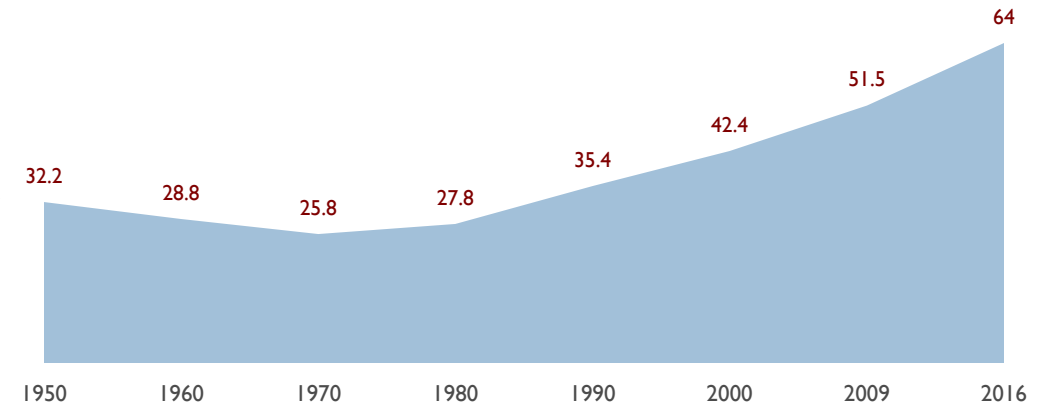
roma04.png,1920,1080,Roma (2018)

caption movies.csv | decksh | pdf ...

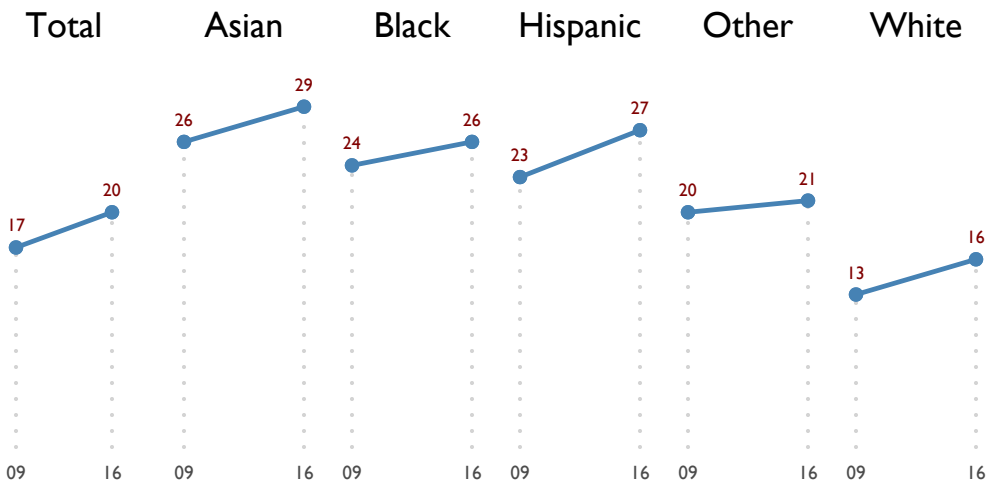
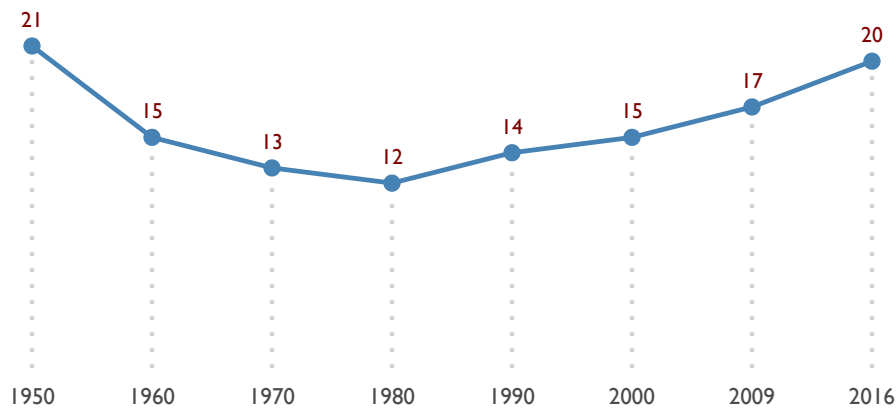
A record 64 million Americans live in multigenerational households

The number and share of Americans living in multigenerational family households have continued to rise, despite improvements in the U.S. economy since the Great Recession. In 2016, a record 64 million people, or 20% of the U.S. population, lived with multiple generations under one roof, according to a new Pew Research Center analysis of census data.

Multigenerational households (millions)

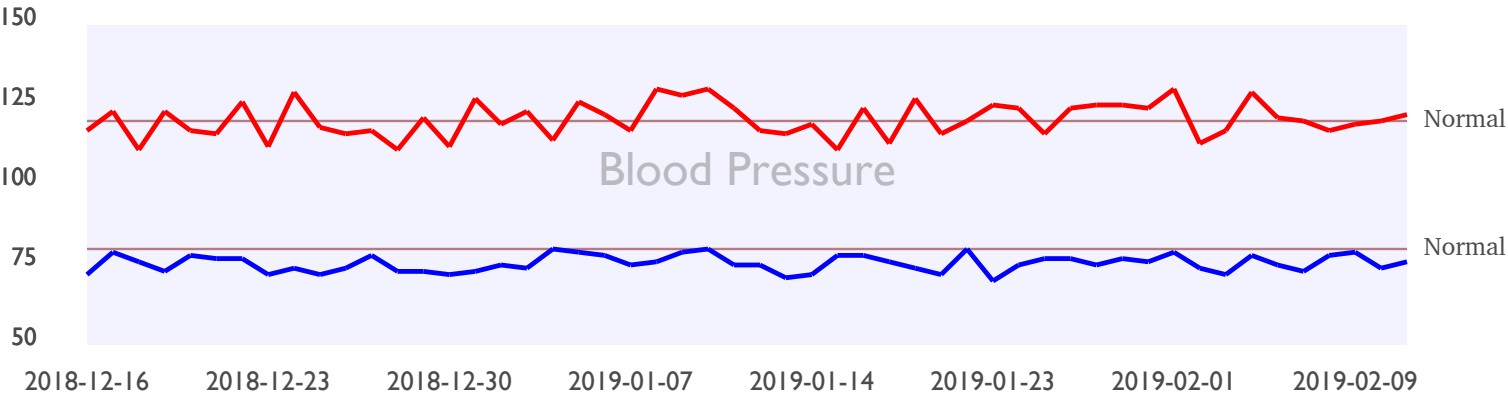


% of Americans in multigenerational households

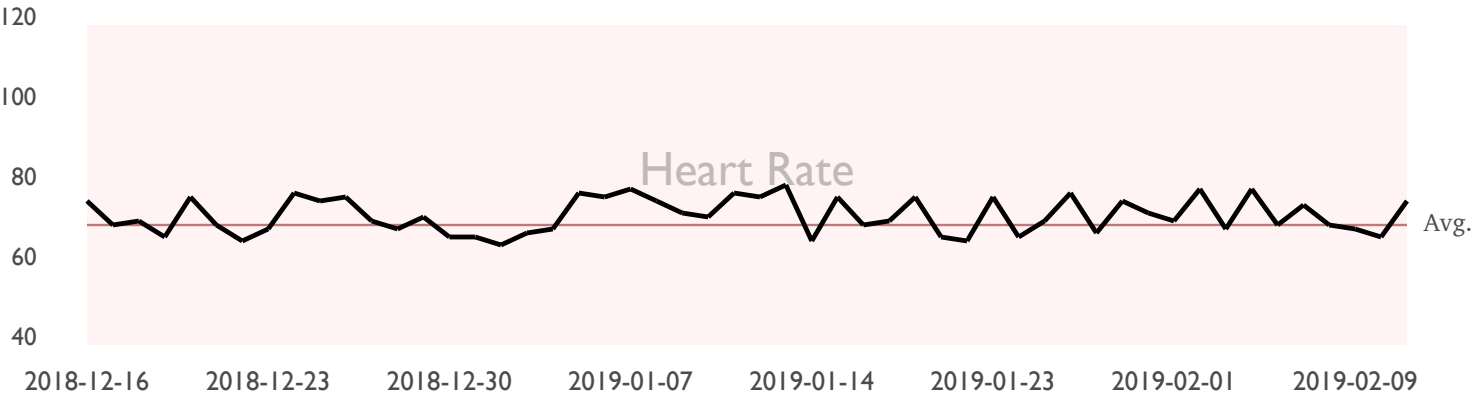


Jane Doe

■ Systolic
■ Diastolic

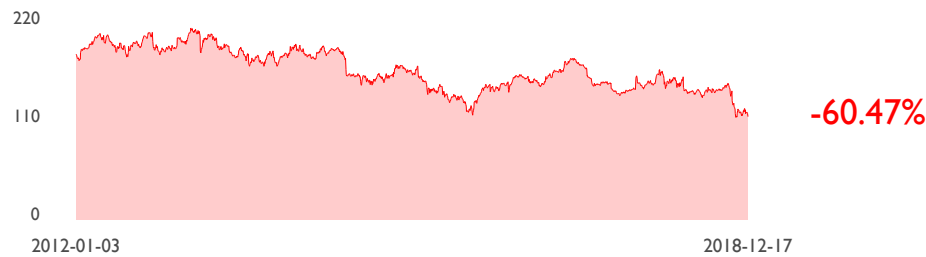


■ Beats/Min





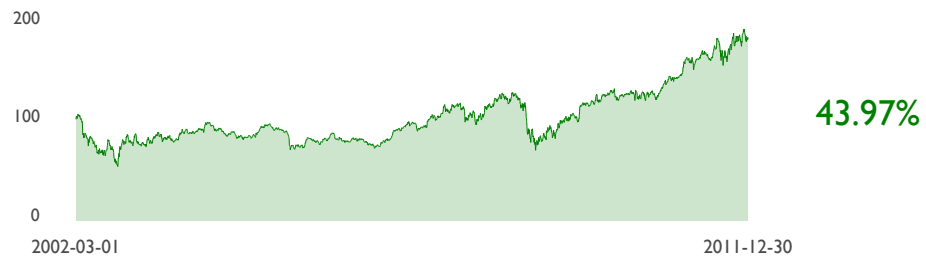
Rometty



The first woman to lead IBM, Rometty shifted IBM away from shrinking businesses such as computers and operating system software, and into higher-growth areas like artificial intelligence. Her tenure has also been met by fierce criticism relating to executive compensation bonuses, layoffs, outsourcing, and presiding over 24 consecutive quarters of revenue decline.



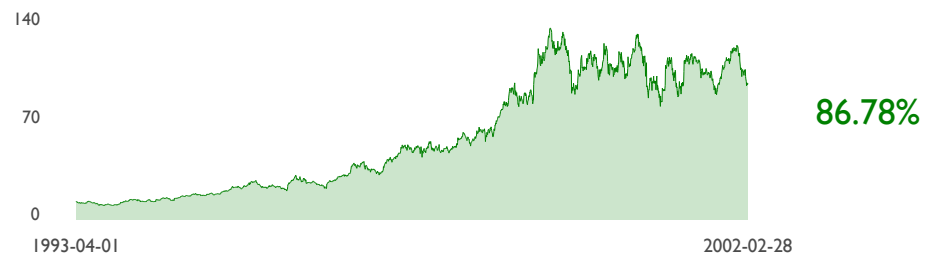
Palmisano



Palmitano's mandate was to move into new unique businesses with high profit margins and potential for innovation. This included purchasing PWC Consulting in 2002, so that IBM could go beyond selling computers and software and help customers use technology to solve business challenges in areas such as marketing, procurement and manufacturing.



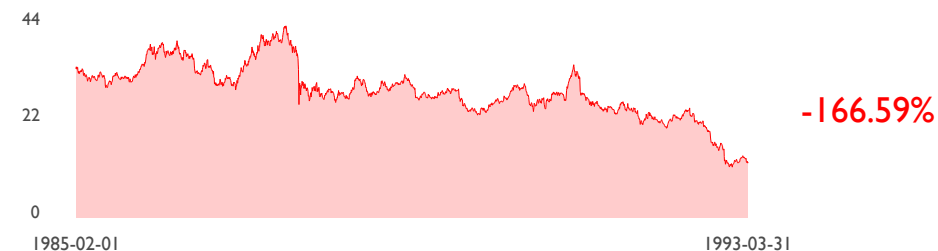
Gerstner



Gerstner's choice to keep the company together was the defining decision of his tenure, as these gave IBM the capabilities to deliver complete IT solutions to customers. Services could be sold as an add-on to companies that had already bought IBM computers, while barely profitable pieces of hardware were used to open the door to more profitable deals.

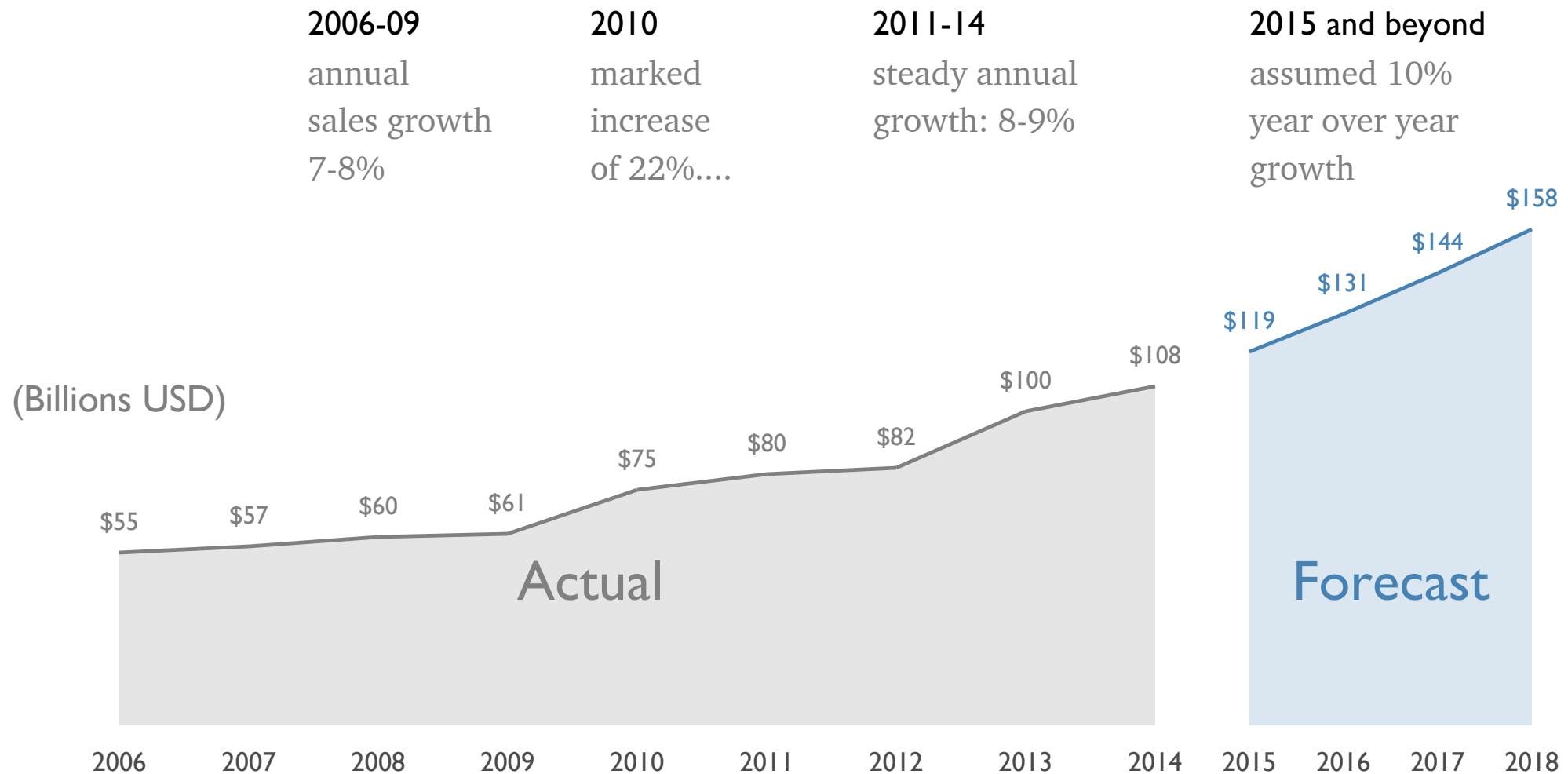


Akers

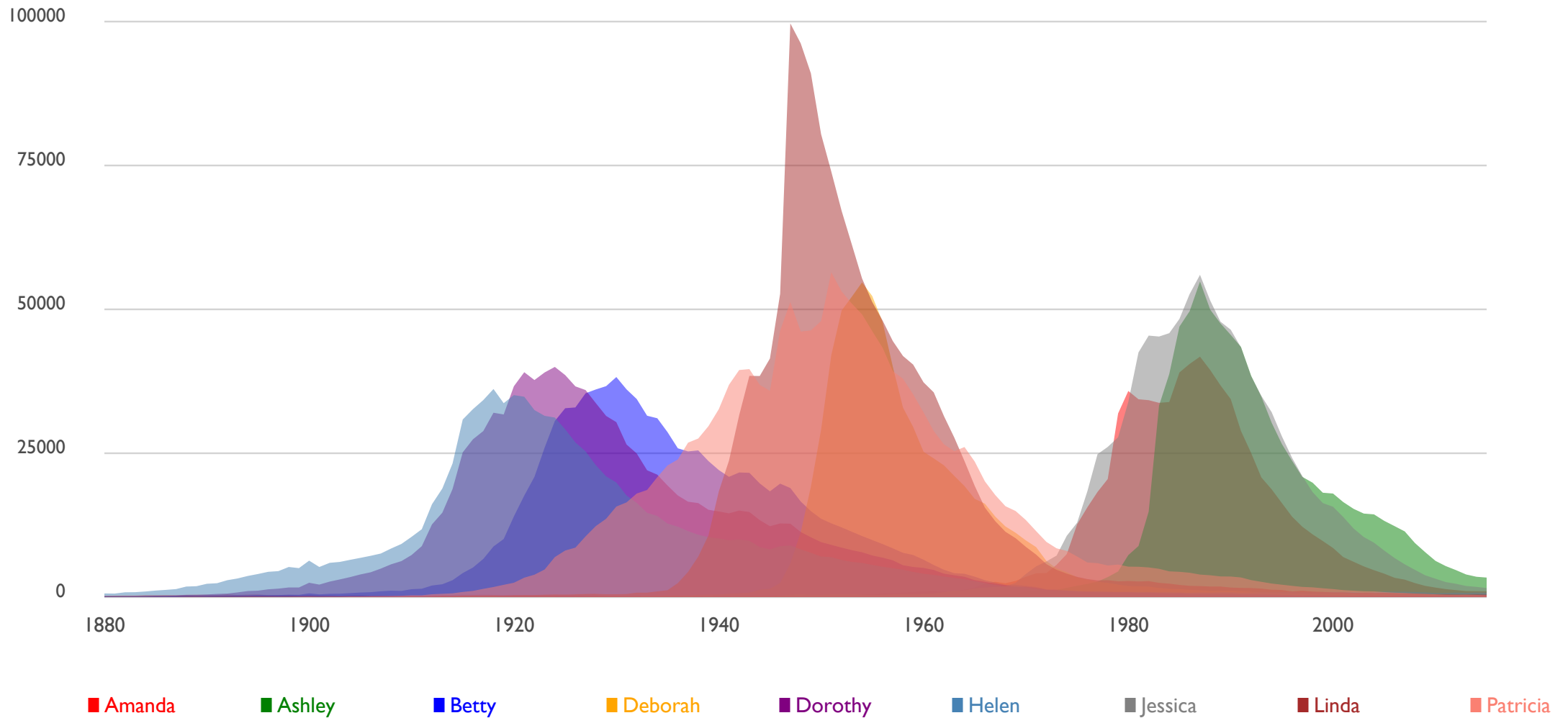


Akers was credited with simplifying the company's bureaucracy to focus more on profits. In a restructuring intended to reverse three years of disappointing performance, he created five new, autonomous organizations responsible for the company's innovation, design and manufacturing. Akers was forced to resign, after the company posted an unprecedented \$5 billion annual loss.

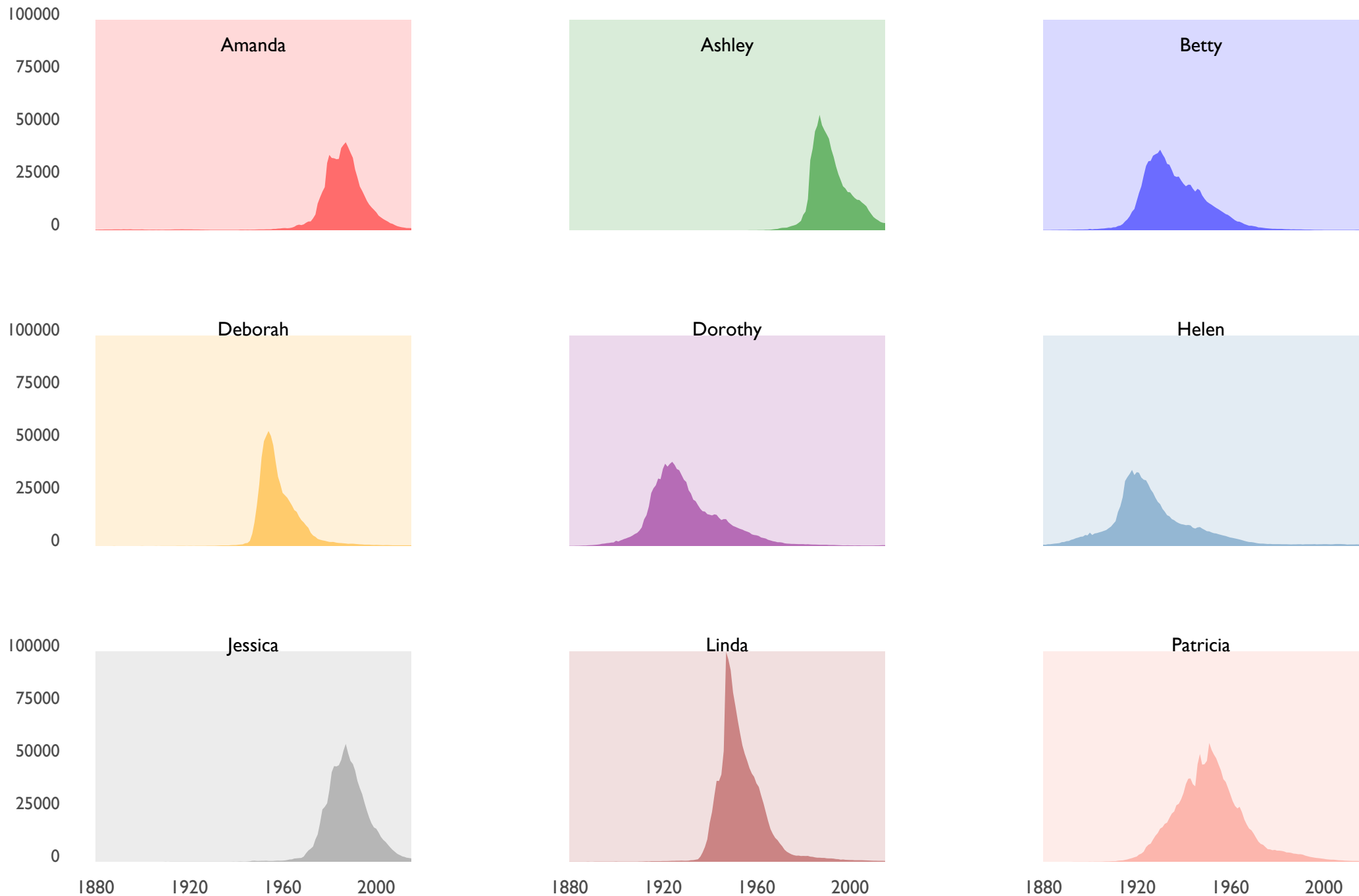
Sales over time



Evolution of Baby Names in the US: 1880-2015

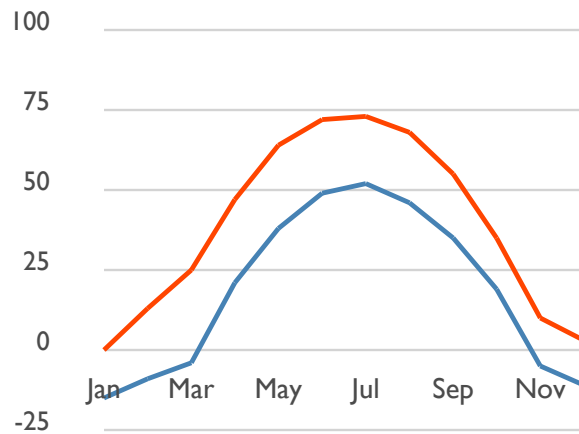


Evolution of Baby Names in the US: 1880-2015

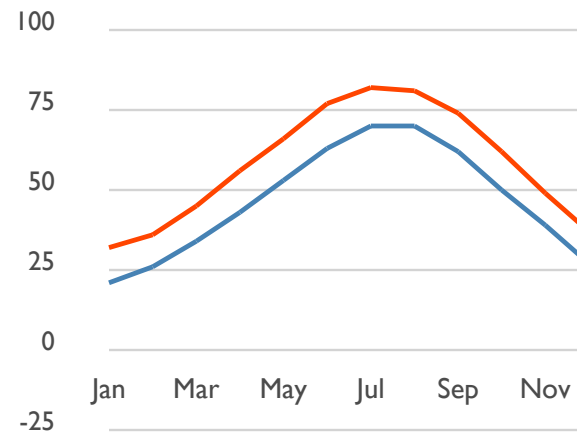


Average High/Low Temperatures (°F)

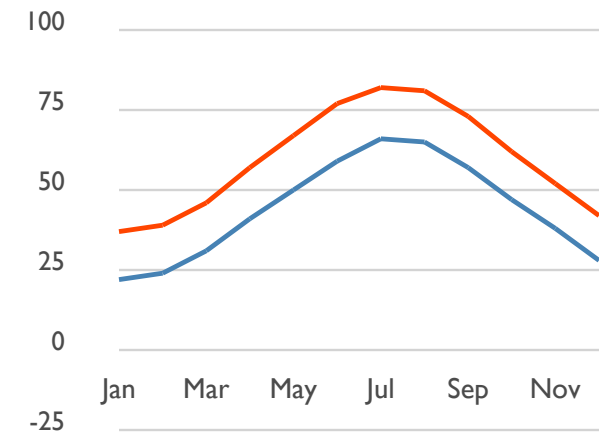
Fairbanks



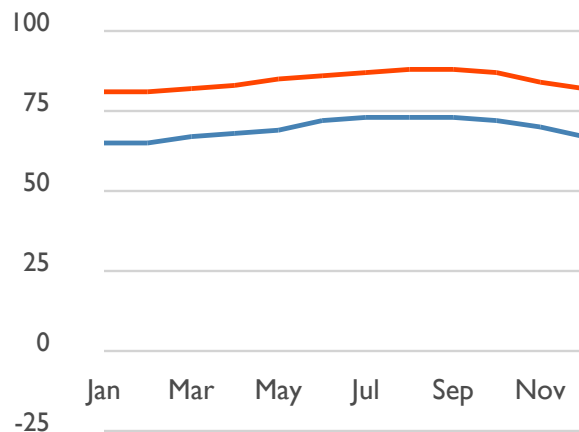
Chicago



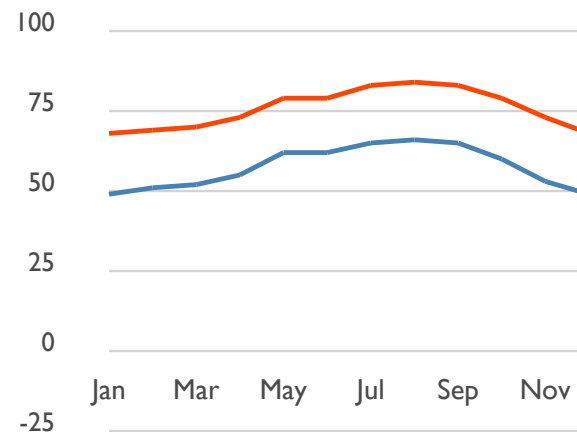
Boston



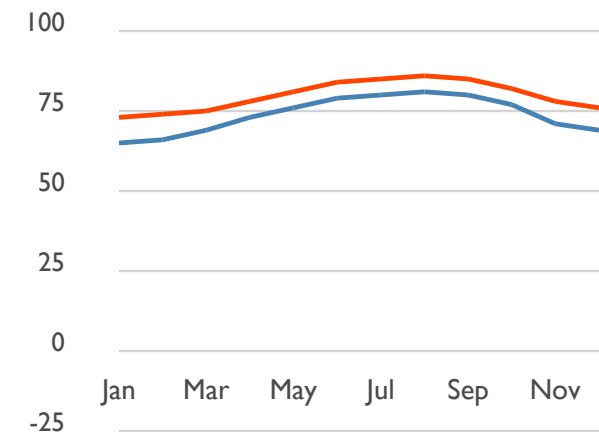
Honolulu



Los Angeles



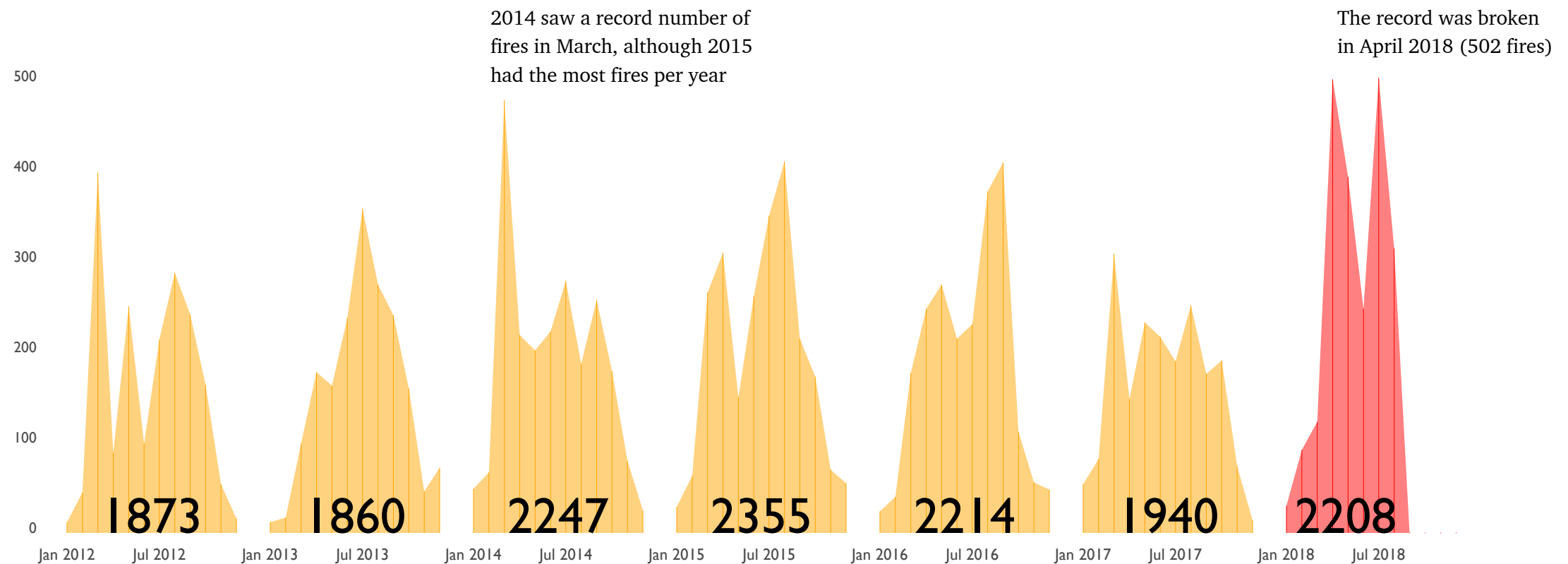
Miami



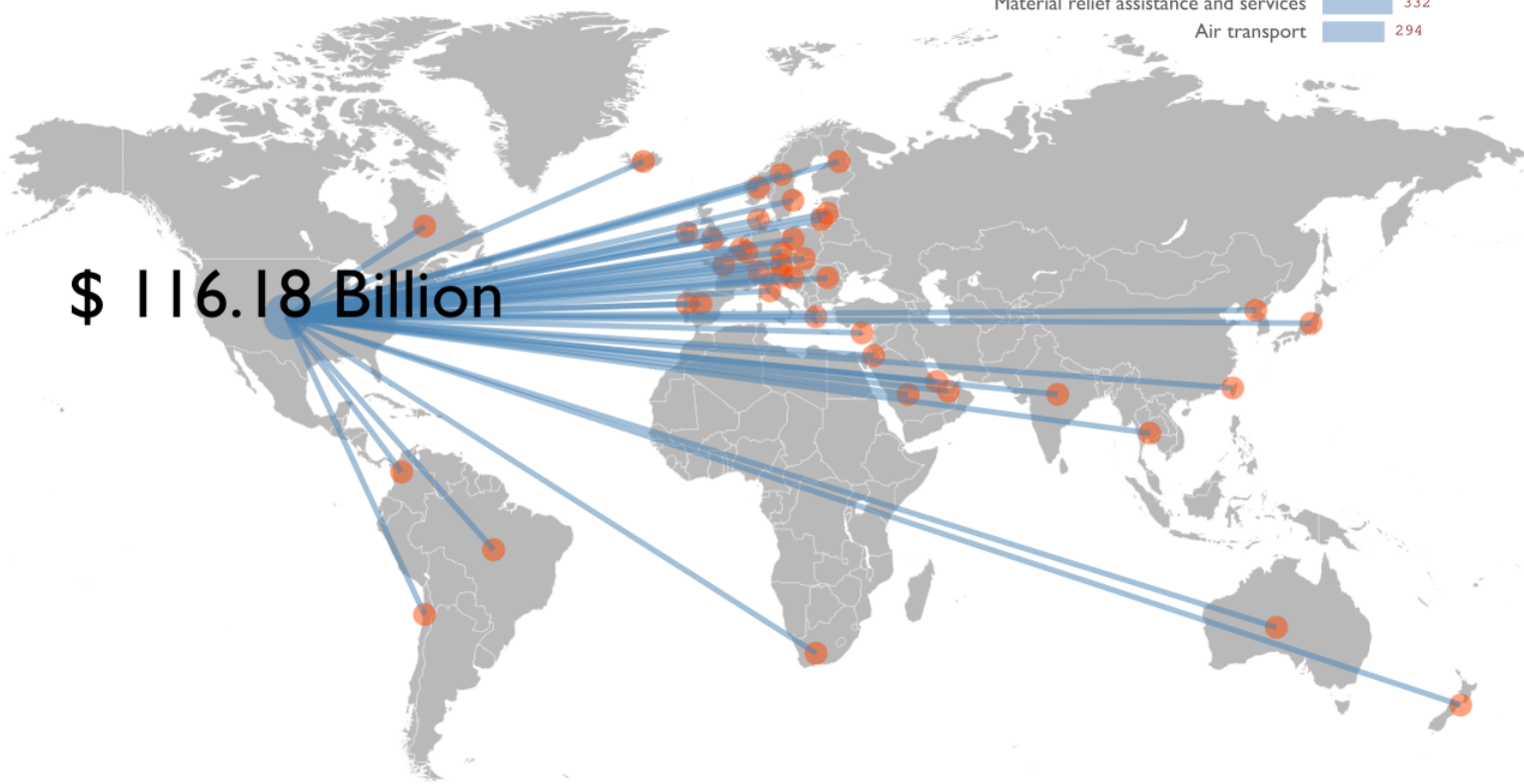
■ Avg. High

■ Avg. Low

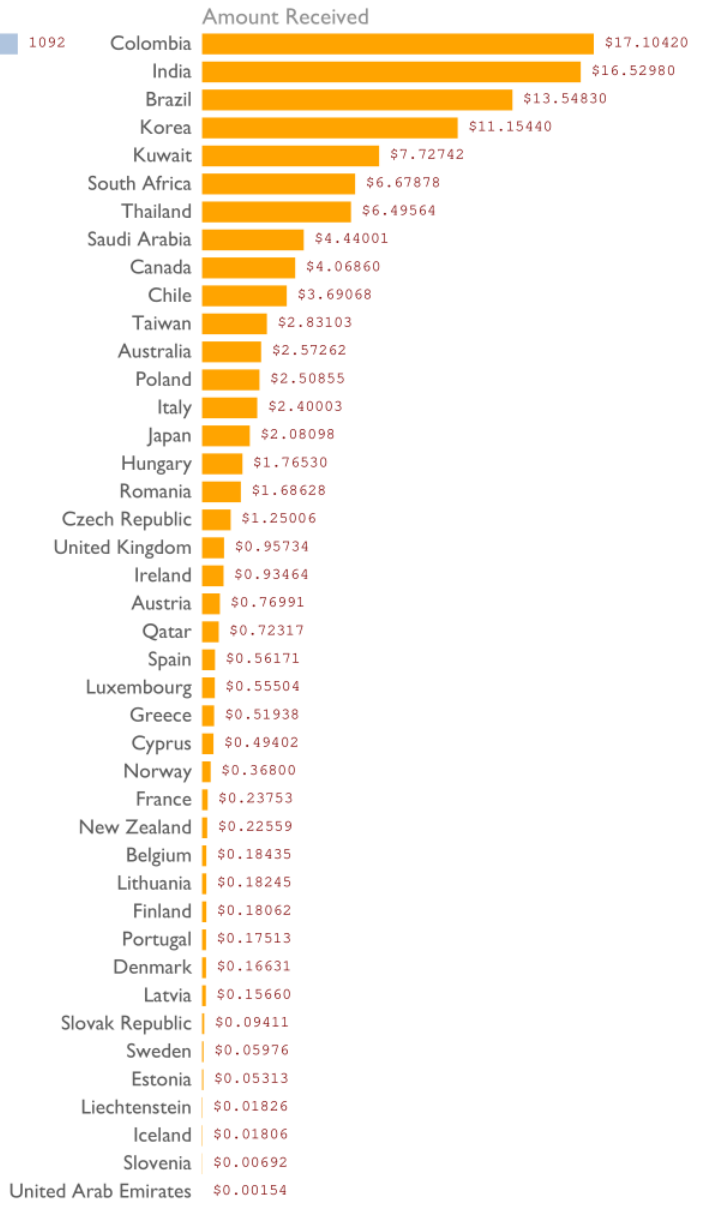
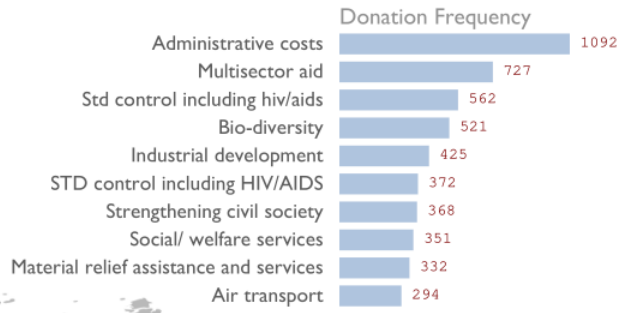
German Wildfires 2012-2018

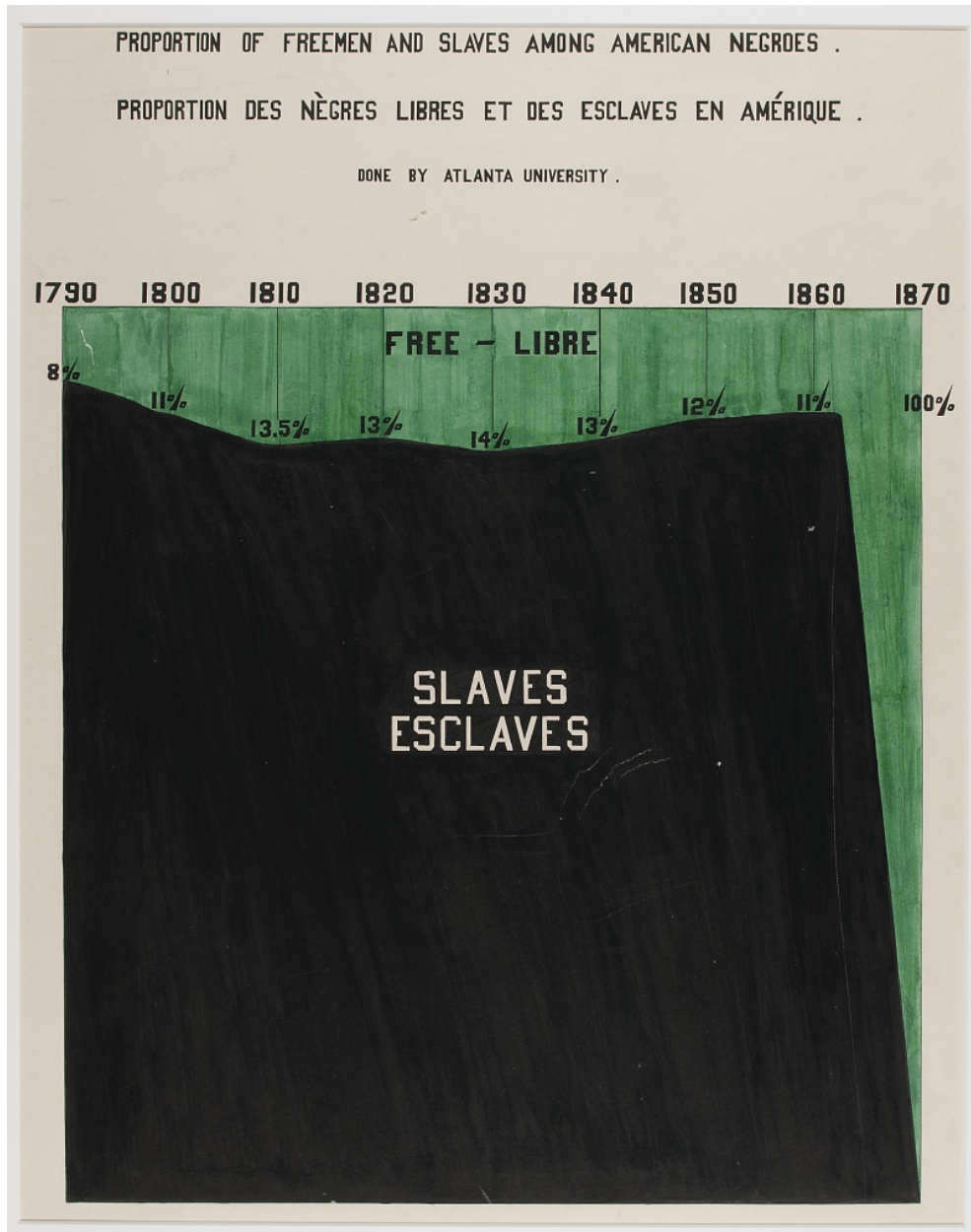


United States



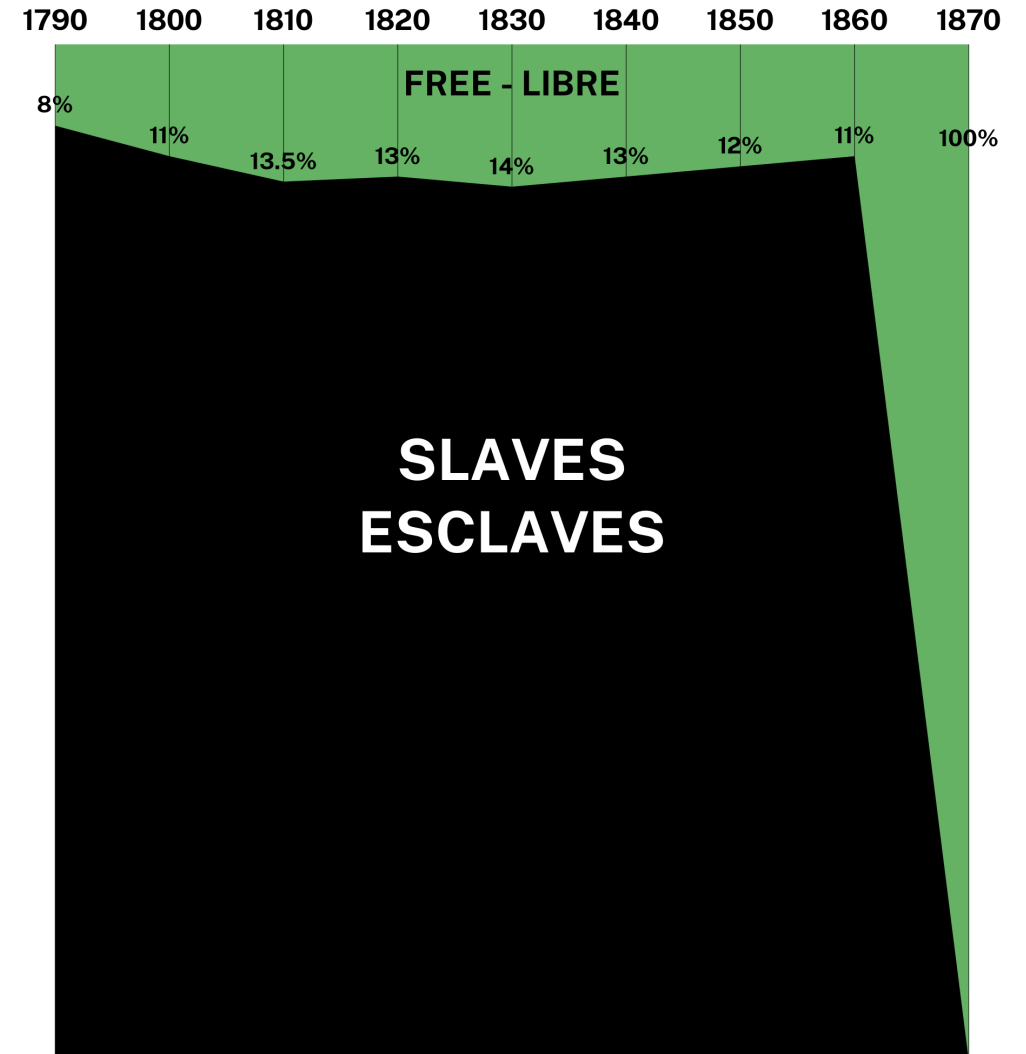
\$ 116.18 Billion





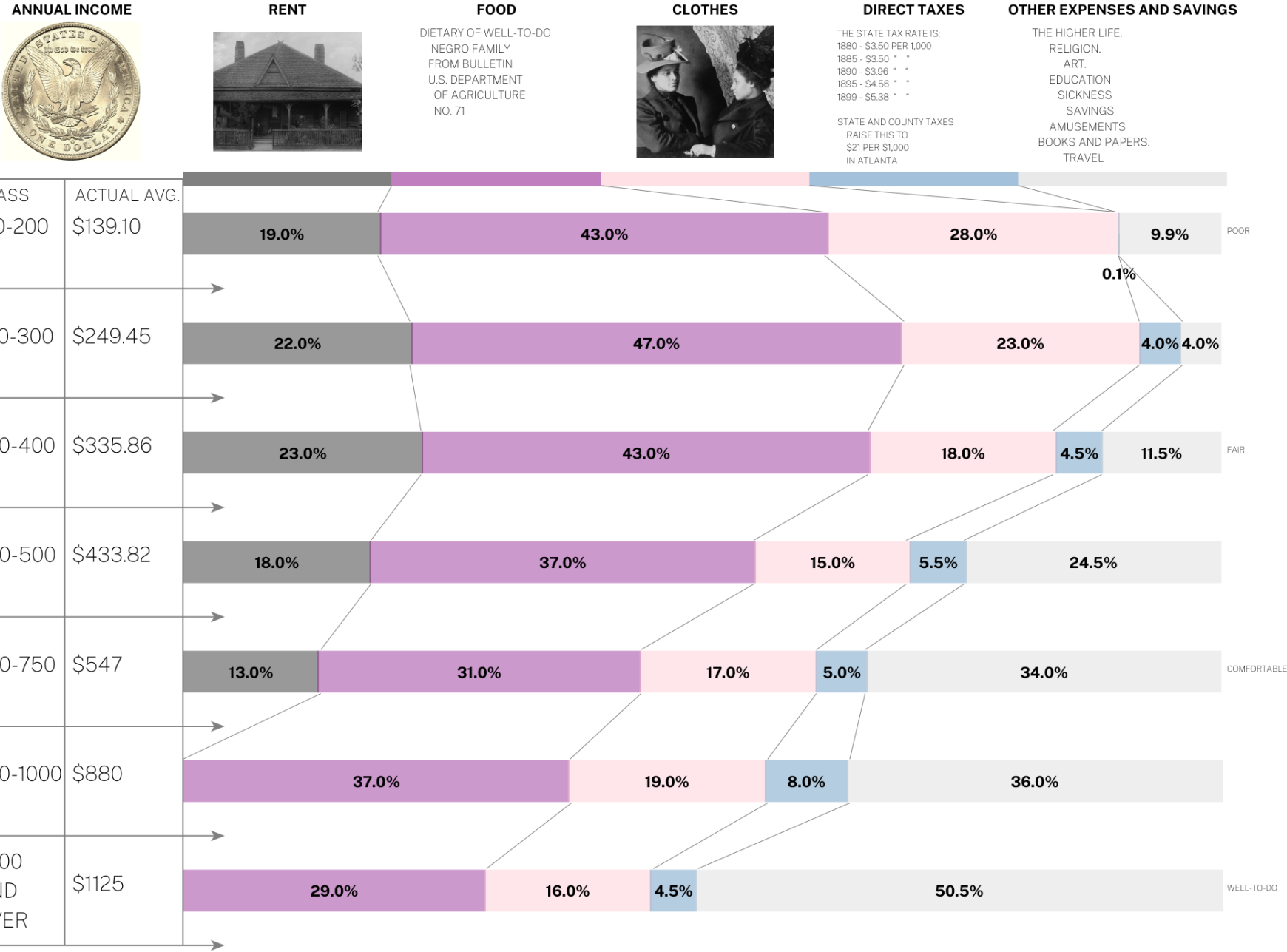
PROPORTION OF FREEMEN AND SLAVES AMONG AMERICAN NEGROES.
PROPORTION DES NÈGRES LIBRES ET DES ESCLAVES EN AMÉRIQUE.

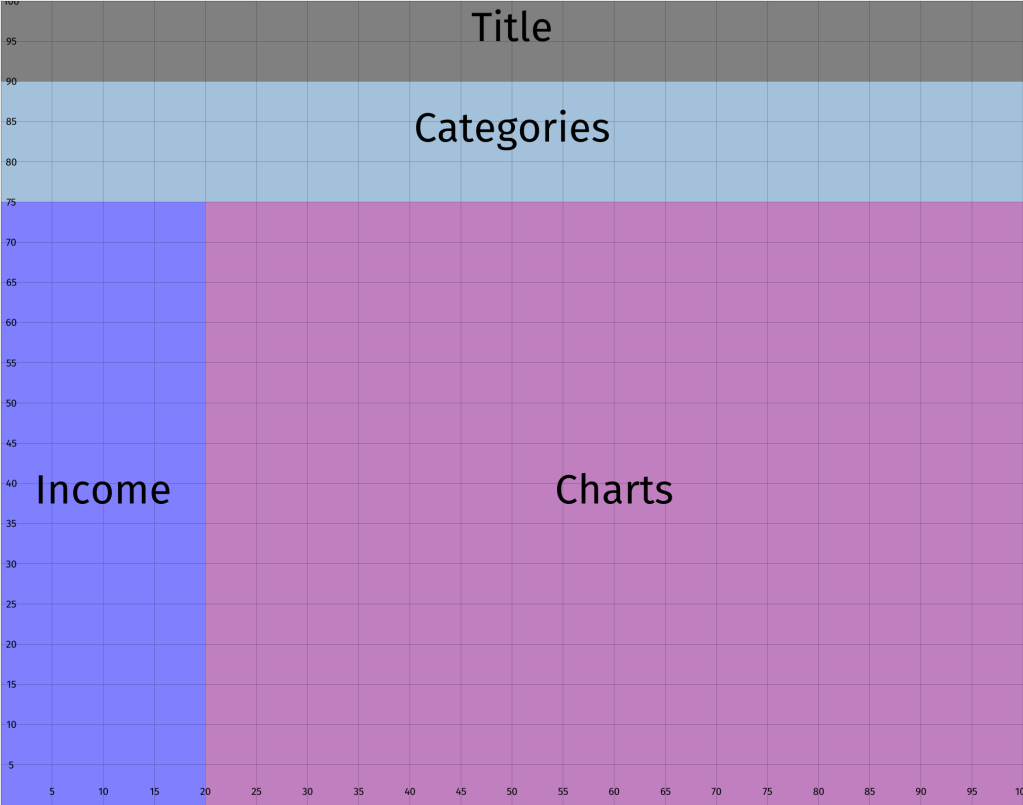
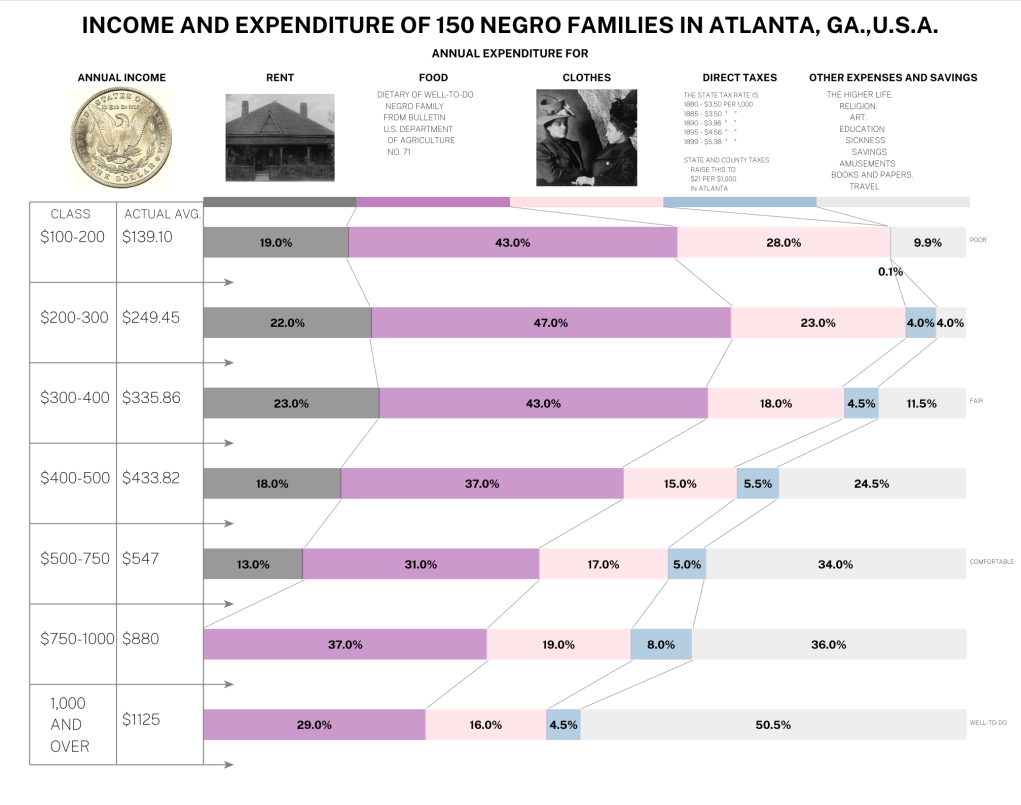
DONE BY ATLANTA UNIVERSITY.



INCOME AND EXPENDITURE OF 150 NEGRO FAMILIES IN ATLANTA, GA.,U.S.A.

ANNUAL EXPENDITURE FOR





go get it

decksh

github.com/ajstarks/decksh

dchart

github.com/ajstarks/dchart

pdfdeck

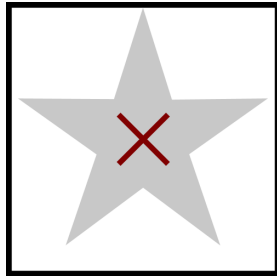
github.com/ajstarks/deck/cmd/pdfdeck

examples

github.com/ajstarks/deckviz

fonts

github.com/ajstarks/deckfonts



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github.com/ajstarks



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